

Yeh bada hi mazedar chapter hai jo mains or Adv dono mein khoob poocha jata hai..... Bas difference itna hai ab tak jo humne physics padhi hai uski analysis humne body ko particle maan kar kiya hai actual shape and size ko hum COM and rotation mein consider karte hai let's start.



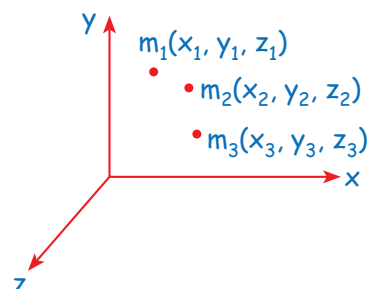
Centre of mass is an imaginary point of any extended body or a system of particles which will perform same motion as a particle of same mass would have performed under applications of same external conditions.

Waise toh COM ki actual feel tumhe dheere-dheere aayegi lekin starting ke liye aise samjho agar tumne kisi ek cricket ball ko spin karane ke liye ghuma kar feka toh ball ke surface ka har ek point bada ajeeb sa utpatang motion karega infect andar wale point bhi but in sabhi mein ball ka centre ka motion bilkul parabolic hoga jaisa humne kinematics mein padha tha just like motion of a particle thrown in projectile isi special point ko COM kehte hai.



Pehle hi ball pe sixer

LOCATION OF C.O.M FOR DISCRETE PARTICLE SYSTEM

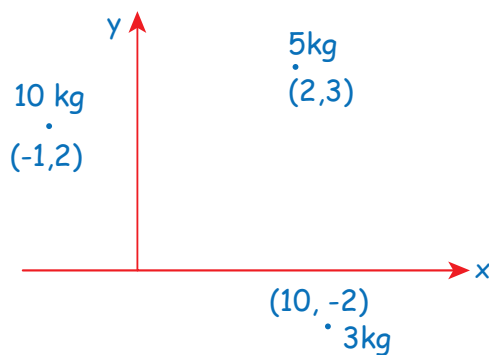


$$x_{\text{com}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum m_i x_i}{\sum m_i}$$

$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3} = \frac{\sum m_i y_i}{\sum m_i}$$

$$z_{\text{com}} = \frac{m_1 z_1 + m_2 z_2 + m_3 z_3}{m_1 + m_2 + m_3} = \frac{\sum m_i z_i}{\sum m_i}$$

Q. Find location of COM of following system.

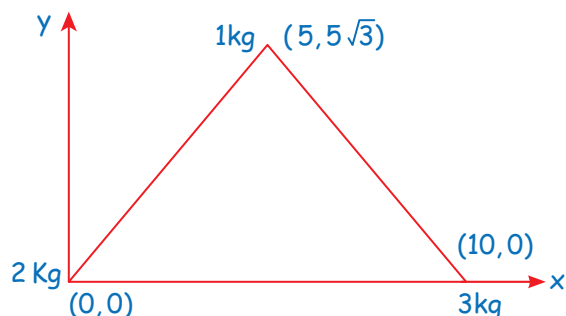


$$\text{Sol. } x_{\text{com}} = \frac{(10 \times (-1)) + 5 \times 2 + 3 \times 10}{10 + 8} = \frac{30}{18}$$

$$y_{\text{com}} = \frac{20 + 15 - 6}{18} = \frac{29}{18}$$

$$\text{com} = \left(\frac{30}{18}, \frac{29}{18} \right)$$

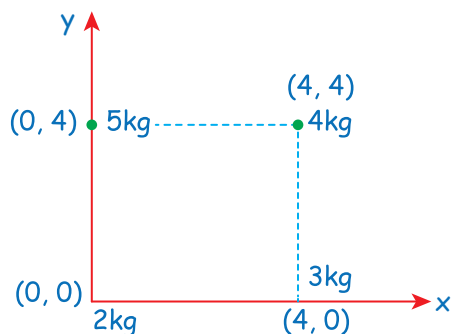
Q. Find location of COM of following system.



Sol. (f) $x_{\text{com}} = \frac{0 + 5 + 30}{6} = \frac{35}{6}$

$y_{\text{com}} = \frac{0 + 5\sqrt{3} + 0}{6} = \frac{5\sqrt{3}}{6}$

Q. Find COM.



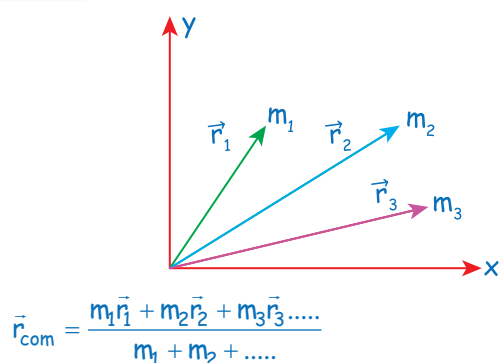
Sol. $x_{\text{com}} = \frac{(2 \times 0) + (3 \times 4) + (4 \times 4) + (5 \times 0)}{5 + 4 + 2 + 3}$

$= \frac{12 + 16}{14} = 2$

$y_{\text{com}} = \frac{(2 \times 0) + (3 \times 0) + (4 \times 4) + (5 \times 4)}{5 + 4 + 2 + 3}$

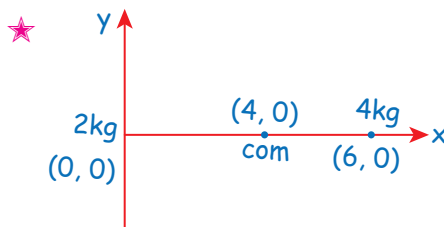
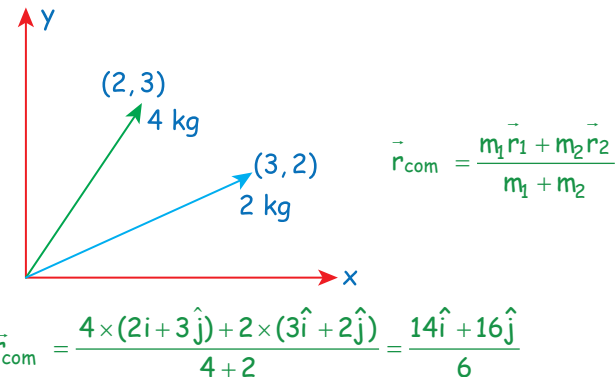
$\frac{36}{14} = \frac{18}{7}$

COM OF DISCRETE PARTICLE (IN VECTOR FORM)



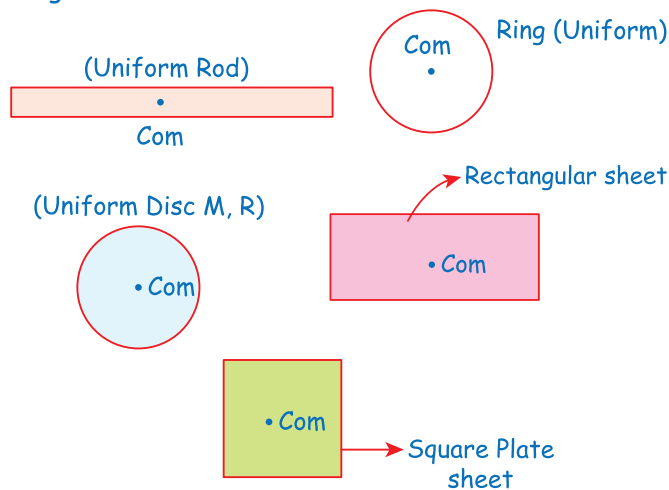
Centre of Mass

Q. Find location of COM of following system.



★ COM of two particle system will lie on the line joining the two particle system will be close to heavier mass.

★ COM of uniform symmetric body lie on their geometrical centre.

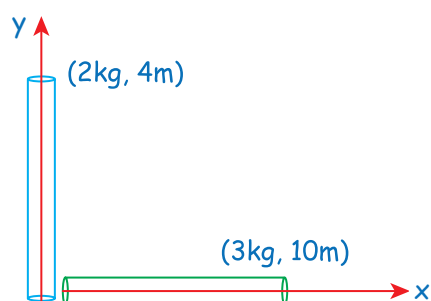


COM OF CONTINUOUS MASS SYSTEM

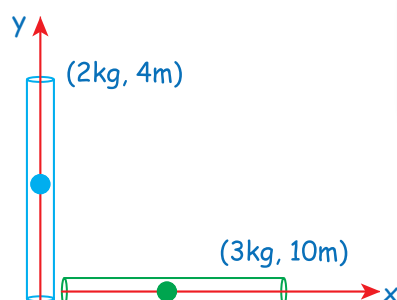
$x_{\text{com}} = \frac{\int x \, dm}{\int dm}$ x-coordinate of COM of dm

$y_{\text{com}} = \frac{\int y \, dm}{\int dm}$ y-coordinate of COM of dm

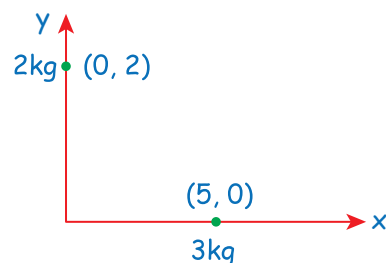
Q. Find location of COM if both rod are uniform.



Sol.



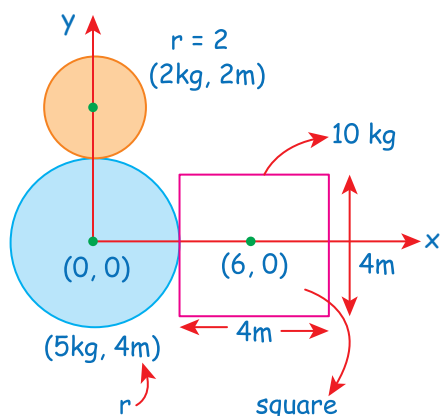
SKC
body hatao
bindu banao



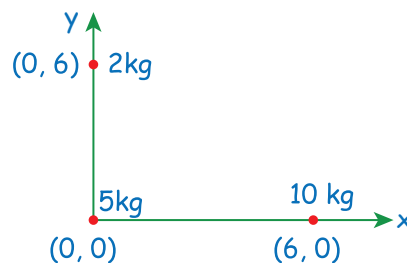
$$x_{\text{com}} = \frac{(3 \times 5) + (2 \times 0)}{2 + 3} = 3$$

$$y_{\text{com}} = \frac{0 + 2 \times 2}{2 + 3} = \left(\frac{4}{5}\right)$$

Q. Find location of COM of following system.



Sol.

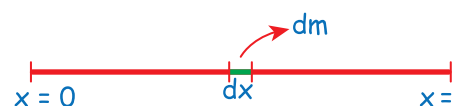


$$x_{\text{com}} = \frac{0 + 0 + 60}{(m_1 + m_2 + m_3)17} = \frac{60}{17}$$

$$y_{\text{com}} = \frac{12 + 0 + 0}{17} = \frac{12}{17}$$

MASS DISTRIBUTION

(1) Linear mass density λ = mass per unit length



$$dm = \lambda dx$$

ye dx length

ka chhota sa mass dm hai

$$\lambda = \frac{dm}{dx}$$

$$M_{\text{total}} = \int dm = \int_{x=0}^{x=l} \lambda dx.$$

If λ is uniform

$$m_{\text{Total}} = \lambda \int_0^l dx$$

$$m = \lambda(l - 0)$$

$$\lambda = \frac{m}{L}$$

(2) Area mass density σ (sheet, Disc)

σ = mass per unit Area

$$\sigma = \frac{dm}{dA}$$

$$dm = \sigma dA$$

$$m_{\text{total}} = \int \sigma dA$$

If σ is uniform

$$M_{\text{Total}} = \sigma A$$

$$\sigma = \frac{m}{A}$$

(3) Volume mass density (ρ)

ρ = mass per volume

$$\rho = \frac{dm}{dv}$$

$$dm = \rho dv$$

$$m_{\text{total}} = m = \int \rho dv$$

If $\rho \rightarrow$ Uniform

$$m_{\text{Total}} = \rho \text{ vol}^n$$

$$\rho = \frac{\text{mass}}{\text{volume}}$$

Q. A rod of length 10 m is placed on x-axis such that it's one end at origin and another at (10, 0) if linear mass density λ varies as $\lambda = 4x^2$.

Find (a) total mass of rod (b) Location of COM.

Sol. (a) $M_{\text{Total}} = \int \lambda dx = \int_0^{10} 4x^2 dx = \frac{4000}{3}$

(b) $x_{\text{com}} = \frac{\int x dm}{\int dm} = \frac{\int x \lambda dx}{m_{\text{total}}}$

$x_{\text{com}} = \frac{\int_0^{10} x \cdot 4x^2 dx}{\frac{4000}{3}} = \frac{4 \times \frac{10000}{4}}{\frac{4000}{3}} = \frac{15}{2}$

Q. Repeat the above problem if length of rod is l and $\lambda = \lambda_0 x$

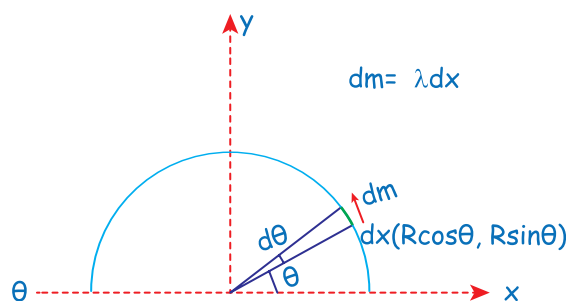
Sol. $M_{\text{Total}} = \int \lambda dx = \int_0^l \lambda_0 x dx = \frac{\lambda_0 l^2}{2}$

$x_{\text{rod}} = \frac{\int x dm}{\int dm}$

$x_{\text{com}} = \frac{\int_0^l x \lambda dx}{\int_0^l \lambda dx} = \frac{\int_0^l x \lambda_0 x dx}{\int_0^l \lambda_0 x dx} = \frac{\frac{l^2}{3}}{\frac{l^2}{2}} = \frac{2l}{3}$

COM OF SEMI RING (M, R)

$x_{\text{com}} = 0$



$\Rightarrow dx = R d\theta \Rightarrow dm = \lambda R d\theta$

$y_{\text{com}} = \frac{\int y dm}{\int dm} = \frac{\int_0^\pi R \sin \theta \cdot \lambda R d\theta}{\int_0^\pi \lambda R d\theta}$

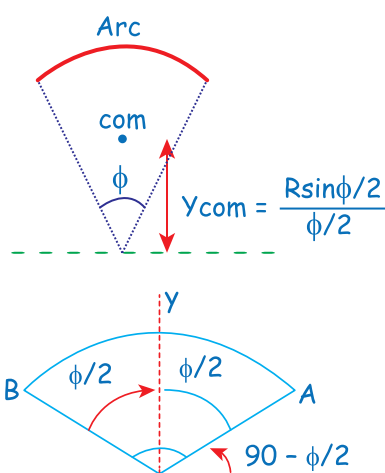
$= \frac{\lambda R^2 \cdot \int_0^\pi \sin \theta d\theta}{\lambda R \int_0^\pi d\theta} = \frac{R(-\cos \theta)|_0^\pi}{(\pi - 0)}$

$\frac{R}{\pi} (\cos \pi - \cos 0) = \frac{-R}{\pi} (-1 - 1) = \frac{2R}{\pi}$

θ par jake chhota sa angle $d\theta$ pakra jiske samne ki arc length dx hai such that $d\theta = dx/R \Rightarrow dx = R d\theta$ (using $\theta = \text{arc}/\text{radius}$)



COM OF CIRCULAR ARC OF (RADIUS R)



Iska derivation almost same as for semi ring bas only limit change. Agar ham dhyan se dekhne toh $(\pi/2 - \pi/2)$ se $(\pi/2 + \pi/2)$ tak jana h



$y_{\text{com}} = \frac{\int y dm}{\int dm} = \frac{\int_A^B R \sin \theta \lambda R d\theta}{\int_A^B \lambda R d\theta}$

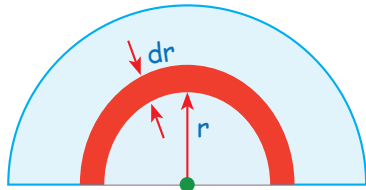
$= \frac{\int_{\pi/2 - \phi/2}^{\pi/2 + \phi/2} R \sin \theta \lambda R d\theta}{\int_{\pi/2 - \phi/2}^{\pi/2 + \phi/2} \lambda R d\theta} = \frac{R \sin(\phi/2)}{\phi/2}$

Q. Find location of com.



Sol. $\phi = 120^\circ \Rightarrow y_{\text{com}} = \frac{R \sin(\pi/3)}{\pi/3} = \frac{3\sqrt{3} R}{2\pi}$

COM OF SEMI DISC



Yahan humne r par jaake dr thickness ka ek chhota sa semi ring pakda jiska area we can write

$$dA = \pi r dr$$

$$dm = \sigma dA = \sigma \pi r dr$$

This is very very very imp concept

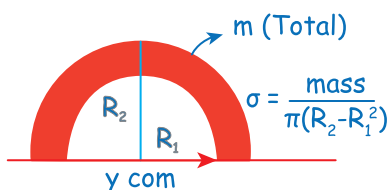


$$x_{com} = 0$$

$$y_{com} = \frac{\int y dm}{\int dm} = \frac{\int_0^R \frac{2r}{\pi} \cdot \sigma \pi r dr}{\int_0^R \sigma \pi r dr}$$

$$= \frac{2 \int_0^R r^2 dr}{\pi \int_0^R r dr} = \frac{2}{\pi} \frac{R^3}{\frac{R^2}{2}} = \boxed{\frac{4R}{3\pi}}$$

Q. Find com to following strip.



Sol. $y_{com} = \frac{\int_{R_1}^{R_2} \frac{2r}{\pi} \pi r dr \sigma}{\int_{R_1}^{R_2} \pi r dr \sigma}$

$$= \frac{4(R_1^2 + R_1 R_2 + R_2^2)}{3\pi(R_1 + R_2)}$$

Bhai yeh bhi toh bilkul semi disc jaisa hi ques hai bas limit R_1 se R_2 kar do... Bas itna hi... hai.

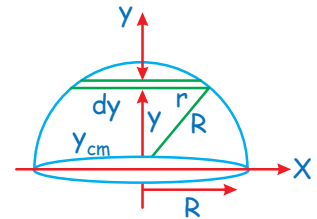
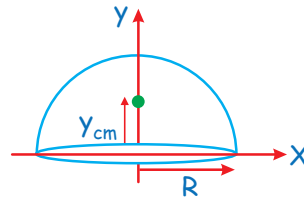


Solid hemisphere, hollow hemisphere and solid cone ke COM find out karne ka derivation neeche attach kar raha hoon jo ki important nahi hai bas inke result yaad rakh lena.

Center of Mass of a Solid Hemisphere

Let's consider an element disc of width dy , mass dm at a distance y from the center of the hemisphere. The radius of this elemental disc will be given as,

$$r = \sqrt{R^2 - y^2}$$



The mass dm of this disc can be given as

$$dm = \frac{3M}{2\pi R^3} \times \pi r^2 dy = \frac{3M}{2R^3} (R^2 - y^2) dy$$

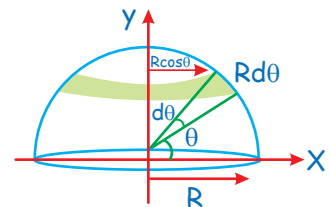
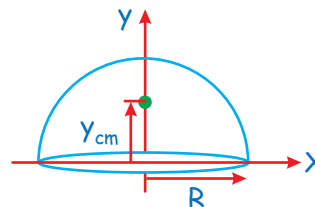
$$y_{cm} = \frac{1}{M} \int_0^R dm y = \frac{1}{M} \int_0^R \left[\frac{3M}{2R^3} (R^2 - y^2) dy \right] y$$

$$\Rightarrow y_{cm} = \frac{3R}{8} \text{ (After solving)}$$

Center of Mass of a Hollow Hemisphere (M, R)

Let's consider an elemental circular strip of angular width $d\theta$ at an angular distance θ from the base of the hemisphere. This strip will have an area.

$$dS = 2\pi R \cos \theta R d\theta$$



$$dm = \frac{M}{2\pi R^2} 2\pi R \cos \theta R d\theta$$

Here y - coordinate of this strip of mass dm can be taken as $R \sin \theta$.

$$y_{cm} = \frac{1}{M} \int_0^{\frac{\pi}{2}} dm R \sin \theta$$

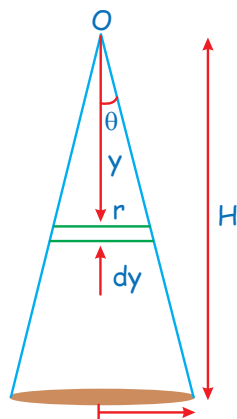
$$= \frac{1}{M} \int_0^{\frac{\pi}{2}} \left(\frac{M}{2\pi R^2} 2\pi R^2 \cos \theta d\theta \right) R \sin \theta$$

$$= R \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta \Rightarrow y_{cm} = \frac{R}{2}$$

Center of Mass of a Solid Cone

A solid cone has mass M , height H and base radius R . Let's consider an elemental disc of width dy and radius r , at a distance y from the apex of the cone. Let the mass of this disc be dm ,

$$dm = \frac{3M}{\pi R^2 H} \times \pi r^2 dy$$



$$y_{cm} = \frac{1}{M} \int_0^H y dm = \frac{1}{M} \int_0^H \left(\frac{3M}{\pi R^2 H} \pi \left(\frac{Ry}{H} \right)^2 dy \right) y$$

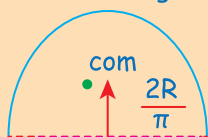
$$= \frac{3}{H^3} \int_0^H y^3 dy = \frac{3H}{4} \text{ (from vertex)}$$



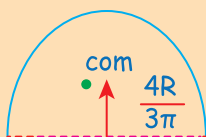
Kaam ka Dabba

★ Exam se pahle ye jarur dekh kr jana

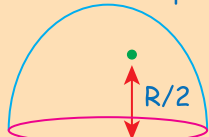
Semi Ring



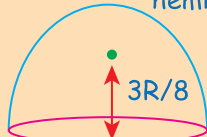
Semi Disc



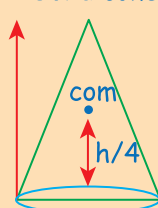
hollow hemisphere



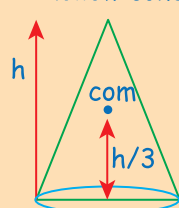
Solid hemisphere



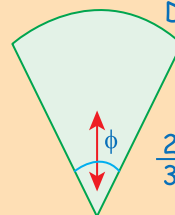
Solid cone



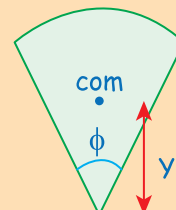
hollow cone



Sector (Part of Disc)

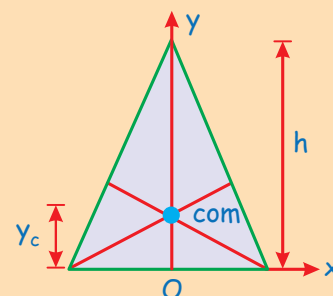


Arc



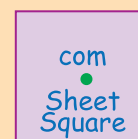
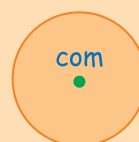
$$y_{com} = \frac{R \sin \phi / 2}{\phi / 2}$$

★ A triangular plate (By qualitative argument)

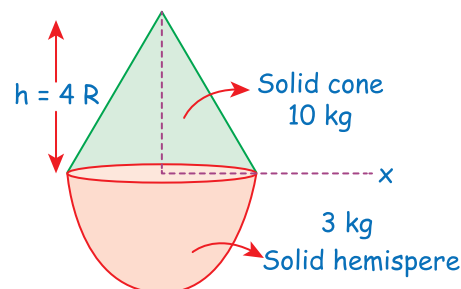


at the centroid: $y_c = \frac{h}{3}$

Ring, Disc, solid and hollow sphere



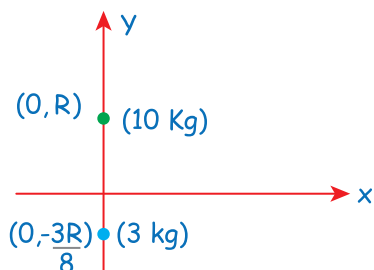
Q. Find location of COM



Aise ques ka ek hi ilaaj, body hatao bindu banao matlab body ke mass ko uske centre of mass par point mass banake solve karlo.



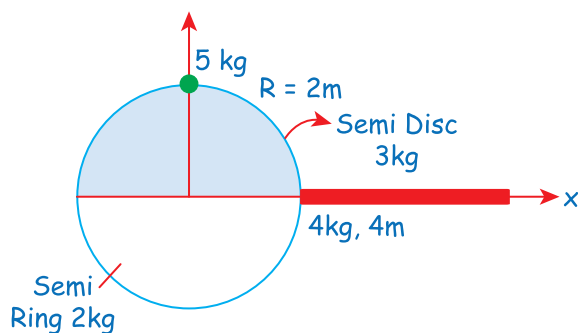
Sol.



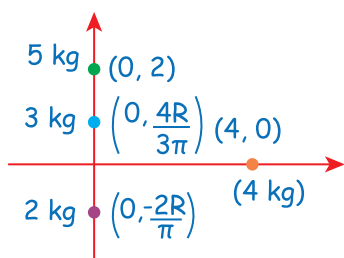
$$x_{\text{com}} = 0$$

$$y_{\text{com}} = \frac{10R + 3 \times (-3R/8)}{10 + 3}$$

Q. In following system we have a point mass of 5 kg, semi disc of 3 kg, semi ring of 2 kg and a uniform rod of length 4 m and mass 4 kg. Find location of COM ($R = 2$ m).



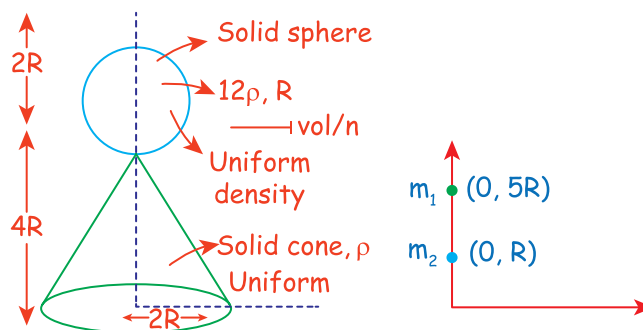
Sol. 4 kg rod has centre at (4, 0)



$$x_{\text{com}} = \frac{0 + 0 + 0 + (4 \times 4)}{14}$$

$$y_{\text{com}} = \frac{5R + \frac{3 \times 4R}{3\pi} - \frac{2 \times 2R}{\pi} + 0}{14} = \frac{5}{7}$$

Q. Find location of COM (density of sphere is 12ρ & that for cone is ρ)



$$\text{Sol. } m_1 = 12\rho \frac{4}{3}\pi R^3$$

$$m_2 = \rho \cdot \frac{1}{3}\pi(2R)^2 4R = \frac{16\pi R^3 \rho}{3}$$

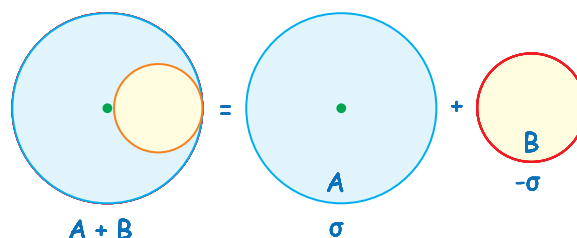
If $m_2 = m$

$$m_1 = 3m$$

$$y_{\text{com}} = \frac{m_1 \times 5R + m_2 \cdot R}{m_1 + m_2}$$

$$= \frac{3m \times 5R + mR}{m + 3m} = \frac{16R}{4} = 4R$$

Q. From a disc (m, R) another disc of Radius is removed as shown in diagram. Find location of centre of mass after removing small Disc.



Man lo ek -ve mass ki disc B, badi vali disc A ke upar raki hai to ye ques wali body 2 disc se milkar bani hai

$$m_A = \sigma \pi R^2 = m (\text{let})$$

$$m_B = -\sigma \pi \left(\frac{R}{2}\right)^2 = \frac{-m}{4}$$

$$\text{Sol. } x_{\text{com}} = \frac{m \times 0 + \frac{-m}{4} \times \frac{R}{2}}{m - \frac{m}{4}} = \frac{-R}{6}$$

$$\text{com} \Rightarrow \left(\frac{-R}{6}, 0\right)$$

MOTION OF COM

Yahaan tak humne location of COM nikaalna seekh liya, ab COM ke motion ko padhenge. Neeche mai COM ke liye bahut important result likh raha hoon, inhe acche se practice kar lo kyunki pura COM inhi par tika hai



Kaam ka Dabba

★ Location of COM = $\vec{r}_{com} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots}{m_1 + m_2 + \dots}$

★ Displacement of COM = $\vec{d}_{com} = \frac{m_1 \vec{d}_1 + m_2 \vec{d}_2 + \dots}{m_1 + m_2 + \dots}$

★ $\vec{v}_{com} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots}{m_1 + m_2} = \frac{\vec{p}_{net}}{m_{total}}$

★ $\vec{p}_{net} = \vec{P} = m_{total} \vec{v}_{com}$ (VVV Important)

★ $\vec{a}_{com} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots}{m_1 + m_2} = \frac{(\vec{F}_{ext})_{net}}{m_{total}}$

★ $(\vec{F}_{net})_{ext} = M_{total} \vec{a}_{com}$ (VVV Important)

★ $\frac{d\vec{p}_{net}}{dt} = (\vec{F}_{net})_{ext} = M_{total} \vec{a}_{com}$

★ If $(\vec{F}_{net})_{ext} = 0 \Rightarrow \vec{a}_{com} = 0$

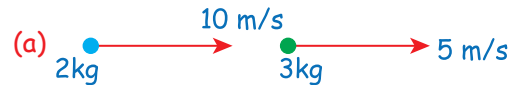
$\vec{v}_{com} = \text{const}$ and momentum of system will conserve

★ If centre of mass is initially at rest and $\sum \vec{F}_{external} = 0$ on centre of mass, then displacement of centre of Mass will be zero. Hence shift in position of centre of mass is zero.

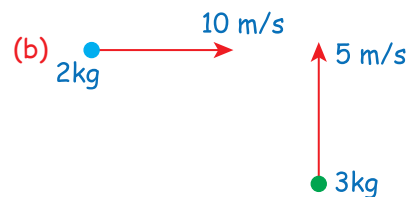
★ If $\sum \vec{F}_{ext}$ is zero on COM then $\vec{v}_{com} = \text{constant}$ but kinetic energy of system may or may not be constant because kinetic energy of the system can also change due to internal forces.

Ye bhut jaruri hai ki aap velocity of COM and acc. of COM nikalna sikhe so let's practice it.

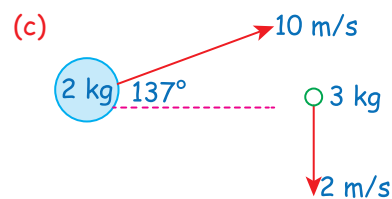
Q. Find velocity of COM in following case.



$$V_{com} = \frac{(2 \times 10) + (3 \times 5)}{2 + 3} = 7 \text{ ms}$$

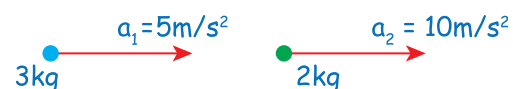


$$V_{com} = \frac{(2 \times 10)\hat{i} + (3 \times 5)\hat{j}}{2 + 3} = 4\hat{i} + 3\hat{j}$$



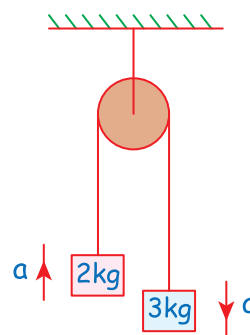
$$V_{com} = \frac{2(8\hat{i} + 6\hat{j}) + 3(-2\hat{j})}{5} = \frac{16\hat{i} - 6\hat{j}}{5}$$

(d) Find acc. of COM.



$$\vec{a}_{com} = \frac{(5 \times 3) + (10 \times 2)}{5} = 7 \text{ m/s}$$

(e) Find acceleration of com.



$$a = \frac{30 - 20}{5} = 2$$

$$a_{com} = \frac{(2 \times 2) + (3 \times (-2))}{2 + 3}$$

$$a_{com} = -\frac{2}{5}$$

If $(\vec{F}_{\text{net}})_{\text{ext}} = 0$

$\vec{P} \rightarrow$ Conserve

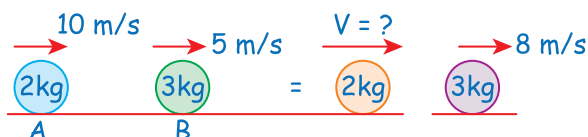
$$\vec{P}_i = \vec{P}_f$$

This is very important concept bas ye dhyaan rakhna aap kisse system maan rahe ho. Jis system par net ext. force 0 hoga uska momentum conserve bole toh const. hoga

Let's practice ques. on this from basic to adv.



Q. Find velocity(v) to 2kg block after collision.

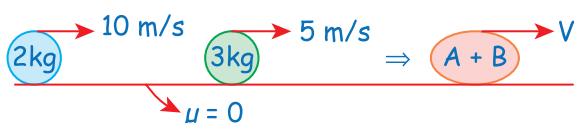


Sol. $P_i = P_f$ $(F_{\text{net ext}})_{A+B} = 0$

$$2 \times 10 + 3 \times 5 = 2V + 3 \times 8$$

$$V = 5.5$$

Q. 2 kg and 3 kg ball strike and stick together. Find velocity of them after chipka-chipki.

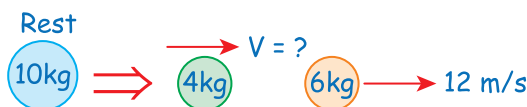


Sol. $P_i = P_f$

$$2 \times 10 + 3 \times 5 = 5 \times V$$

$$V = 7 \text{ m/s}$$

Q. 10 kg block explode into two fragments 4 kg and 6 kg after explosion 6 kg move with velocity 12 m/s along x-axis, answer following parts.



Sol. (1) Velocity of 4 kg block after explosion?

$$\rightarrow P_i = P_f$$

$$0 = 4v + 6 \times 12$$

$$\vec{V} = -18 \rightarrow \text{Backward}$$

(2) V_{com} before and after explosion

$$V_{\text{com}} = 0 = \text{constant}$$

Bcz No external force

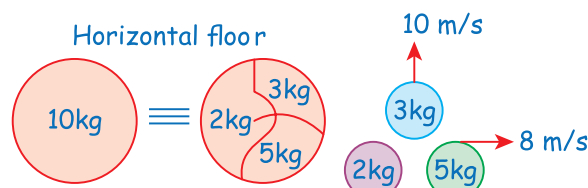
(3) Total energy released in explosion $KE_f - KE_i$

$$KE_i = 0$$

$$KE_f = \frac{1}{2} \times 4 \times 18^2 + \frac{1}{2} \times 6 \times (12)^2$$

$$648 + 432 = 1080$$

Q. 10kg mass explode as shown in figure. Find velocity of (2kg) block.



Sol. Velocity of (2Kg)block

$$P_i = P_f$$

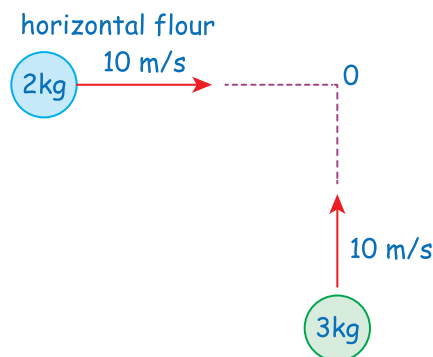
$$0 = 2v + (3 \times 10)\hat{j} + (8 \times 5)\hat{i}$$

$$V = -20\hat{i} - 15\hat{j}$$

Exam mein kabhi-kabhi silly mistake karwane ke liye speed poochh leta hai, we careful! Yahan speed of 2 kg is 25 m/s after explosion



Q. On frictionless horizontal floor two balls are moving as shown in figure.



If both collide at 'O' and stick together Find velocity of (A + B) Just after sticking. and Energy loss in collision.

Sol. Let's after chipka-chipki (sticking) both are moving with velocity V

$$(2 \times 10)\hat{i} + (3 \times 10)\hat{j} = 5\vec{V}$$

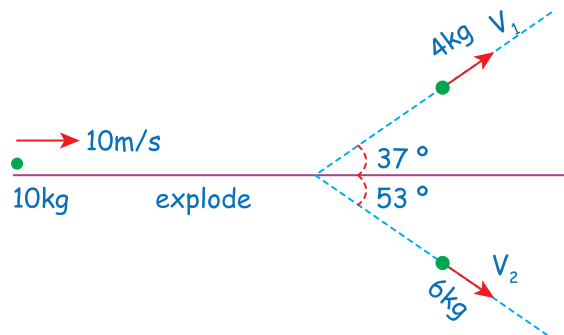
$$\Rightarrow \vec{V} = 4\hat{i} + 6\hat{j}$$

$$(K \cdot E)_i = \frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 10^2 = 250$$

$$(K \cdot E)_f = \frac{1}{2} \times 5 \times (4^2 + 6^2) = 130$$

$$\Delta KE = K \cdot E_f - K \cdot E_i = 130 - 250 = -120 \quad (\text{Loss})$$

Q. On horizontal floor 10 kg shell explode in 2 parts as shown. Find speed of 4 kg and 6 kg.



Sol. Momentum conservation in x

$$P_i = P_f : x \text{ में}$$

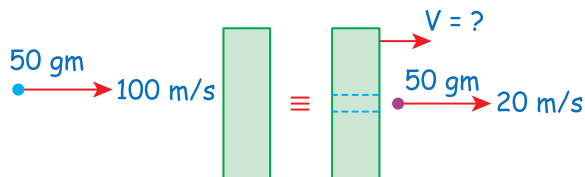
$$10 \times 10 = (4 \times V_1 \cos 37^\circ) + 6(V_2 \cos 53^\circ)$$

momentum conservation in y

$$P_i = P_f : y \text{ में}$$

$$0 = (4V_1 \sin 37^\circ) - (6V_2 \sin 53^\circ) \text{ now you can solve and get the ans.}$$

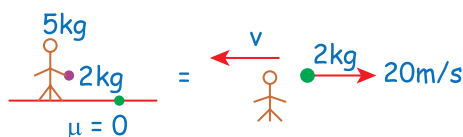
Q. A 50 gm bullet strikes a block and pan through it with velocity 20m/s. Find V.



Sol. Initial momentum $P_i = P_f$ final momentum

$$\frac{50}{1000} \times 100 + 0 = 50v + \frac{50}{1000} \times 20 \Rightarrow V = 0.8 \text{ m/s}$$

Q. Man throws 2kg body as shown in figure. Find velocity of man.



Sol. $(\vec{F}_{\text{net}})_{\text{ext}} = 0$

$$P_i = P_f$$

$$0 = (2 \times 20) - 50 \times V$$

$$V = \frac{40}{50} = \frac{4}{5}$$

Pistol / Bandook / muzzle velocity / recoil velocity wale ques mein same concept hota hai bas aadmi ke haath mein bandook hoti hai. Same concept will use in nuclear physics in 12th class.

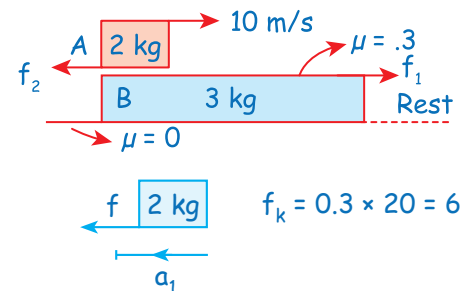


Q. Block A is given velocity 10 m/s as shown above. Answer following questions.

(1) Find when Relative motion stop, or sleeping stop.

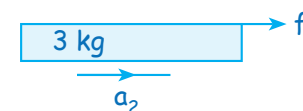
(2) $(V_A \text{ and } V_B)$ sleeping stop or find common V.

Sol.



$$a_1 = \frac{f}{m} = \frac{6}{2}$$

$$a_1 = 3 \text{ (pichhe)}$$



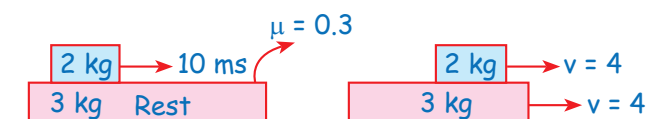
$$a_2 = f_k = \frac{6}{3} = 2$$

Let both have same velocity at time t

$$V = 10 - 3t$$

$$V = 0 + 2t$$

$$t = 2 \text{ sec, } v = 4 \text{ m/s}$$



$$P_i = P_f$$

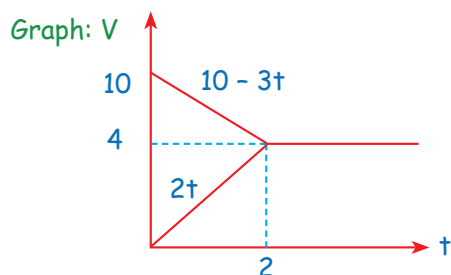
[Momentum conservation]

$$2 \times 10 + 0 = 2V + 3V$$

$$V = 4 \text{ m/s}$$

$$V_{\text{com}} = 0$$

$$V_{\text{com}} = \frac{2 \times 10 + 3 \times 0}{2 + 3} = 4 \text{ m/s same as}$$



$$(KE)_i = \frac{1}{2} \times 2 \times 10^2 + 0 = 100 \text{ J}$$

$$(KE)_f = \frac{1}{2} \times 2 \times 4^2 + \frac{1}{2} \times 3 \times 4^2 = 40 \text{ J}$$

Find the work done by fk:

$$W_g + W_N + W_f = \Delta KE$$

$$0 + 0 + W_f = 40 - 100 = -60$$

SKC

★ WD by kinetic friction can be +ve, -ve or zero but net WD by kinetic friction is always -ve.

★ WD by static friction can be +ve, -ve or zero but net WD by static friction is always zero.

★ WD by internal force are frame independent



Impulse (J)

$$J = \int F \cdot dt$$

★ Since, $F = \frac{dp}{dt}$

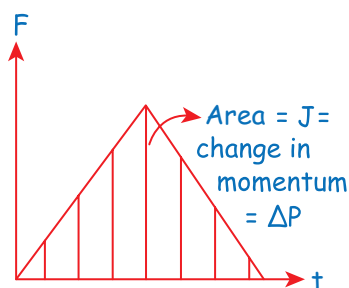
$$\int_{P_i}^{P_f} dp = \int_{t_i}^{t_f} F \cdot dt$$

$$\Delta P = P_f - P_i = \Delta \vec{P} = \int_{t_i}^{t_f} \vec{F} \cdot dt$$

★ $\Delta P = J = \text{Area under } F - t \text{ graph}$

★ $\Delta \vec{P} = \vec{J} = \text{Impulse}$

★ Impulse = change in momentum



Q. Find the impulse due to the force $\vec{F} = a\hat{i} + bt\hat{j}$, where $a = 2 \text{ N}$ and $b = 4 \text{ N/s}$, if this force acts from $t_i = 0$ to $t_f = 0.3 \text{ s}$.

$$\text{Sol. } \vec{I} = \int_{t_i}^{t_f} \vec{F} dt = \int_0^{0.3} (a\hat{i} + bt\hat{j}) dt$$

$$\text{or } \vec{I} = a\hat{i} \int_0^{0.3} dt + b\hat{j} \int_0^{0.3} t dt$$

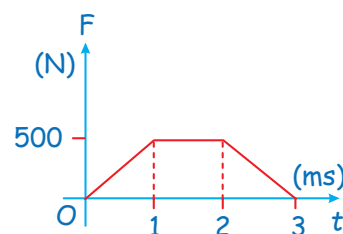
$$\text{or } \vec{I} = at\hat{i} \Big|_0^{0.3} + \frac{bt^2}{2} \hat{j} \Big|_0^{0.3}$$

$$\text{or } \vec{I} = (2)(0.3)\hat{i} + \frac{(4)(0.3)^2}{2} \hat{j}$$

$$\text{thus } \vec{I} = (0.6\hat{i} + 0.18\hat{j}) \text{Ns}$$

Q. Figure shows the variation of force acting on a body with time. Calculate the impulse of this force.

Sol. Impulse of a force is the area under the graph.



Thus,

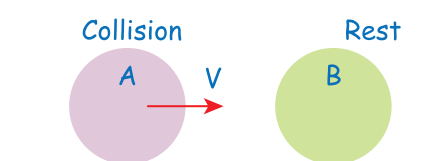
$$I = \left[\frac{1}{2} (500)(1) + (500)(2-1) + \frac{1}{2} (500)(3-2) \right] \times 10^{-3}$$

$$\text{or } I = [250 + 500 + 250] \times 10^{-3}$$

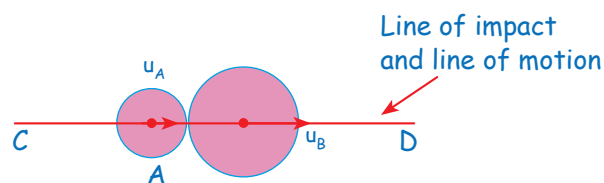
$$\text{or } I = 1 \text{ Ns}$$

COLLISION

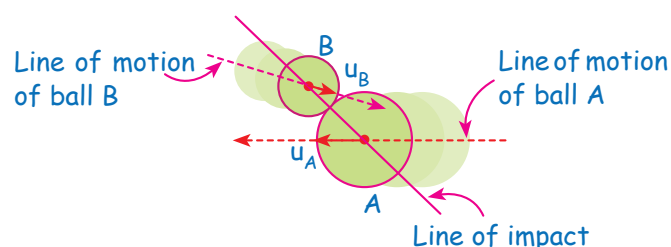
The line passing through the common normal to the surfaces in contact during impact is called line of impact or line of collision (LOI/LOC).



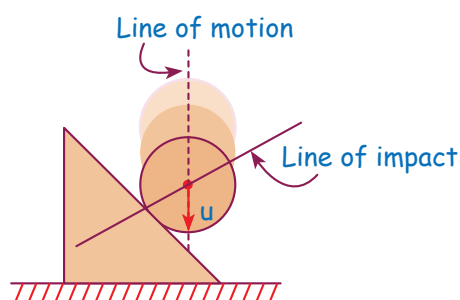
agar body ki velocities before and after the collision LOC ki traf h toh it's called head on collision, else oblique collision.



Head on Collision



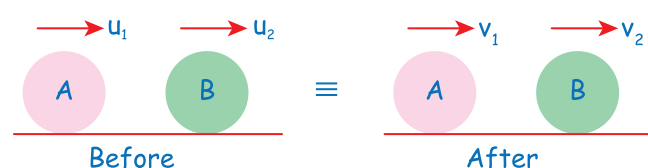
Oblique Collision



Oblique Collision

$$e = \frac{\text{velocity of separation of points of contact along line of impact}}{\text{velocity of approach of points of contact along line of impact}}$$

Head on Collision



$$P_{\text{before}} = P_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\text{Coeff of restitution } e = \frac{v_2 - v_1}{u_1 - u_2}$$

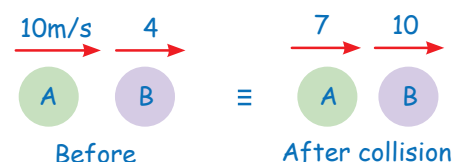
if ($e = 1$) (elastic collision / Perfectly elastic Collision)

if ($0 < e < 1$) (Inelastic collision)

if ($e = 0$) (Perfectly inelastic collision) (stick)

Chalo sabse pehle hum e ki definition lagana seekhte hain.

Q. Find $e = ?$



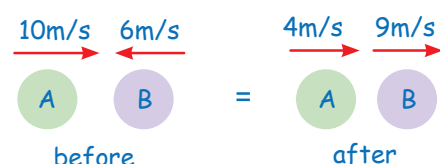
Find $e = ?$

$$\text{Sol. } e = \frac{\text{velocity of sep}}{\text{velocity of app}} = \frac{3}{6} = \frac{1}{2}$$

$$\text{OR } e = \frac{(V_{\text{relative}})_f}{(V_{\text{relative}})_i} = \frac{3}{6} = \frac{1}{2}$$

$$\text{OR } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{10 - 7}{10 - 4} = \frac{1}{2}$$

Q. Find e ?



$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{9 - 4}{10 - (-6)} = \frac{5}{16}$$

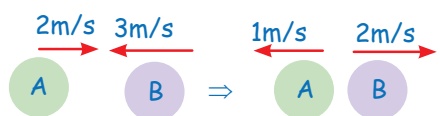
$$\text{Sol. } e = \frac{5}{16}$$

Q. Find e ?



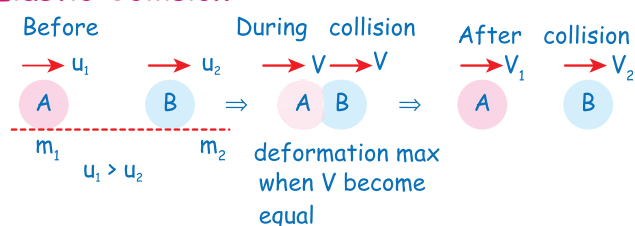
$$\text{Sol. } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{15 - 7}{10 - 0} = \frac{8}{10} = \frac{4}{5}$$

Q. Find e ?



$$\text{Sol. } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{2 - (-1)}{2 - (-3)} = \frac{2 + 1}{5} = \frac{3}{5}$$

Elastic Collision



★ Since, $F_{\text{ext}} = 0$ so, $\vec{P}_i = \vec{P}_f$

★ Momentum $P \rightarrow$ conserve every time

$$m_1 v_1 + m_2 v_2 = m_1 v + m_2 v = m_1 v_1 + m_2 v_2$$

★ (KE) before collision = (KE) after collision

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

★ Overall no energy loss

$$\star e = 1 = \frac{V_2 - V_1}{U_1 - U_2} = \text{coff of restitution}$$

Inelastic Collision

★ $P \rightarrow$ conserve $0 < e < 1$

★ (KE) $\rightarrow$ loss

★ (KE)_{before collision} > (KE)_{after collision}

Perfectly Inelastic Collision

★ $V_1 = V_2$ (KE)_{loss} $\rightarrow$ max

★ $P \rightarrow$ conserve

★ $e = 0$

Dekh bhai, collision ke ques. mein data badal-badal kar khoob ques. pooche jaate hain, isliye is ques. ki practice achhe se kar lena. Yahaan do eqn. important hain - ek momentum conservation aur doosri e ki definition.

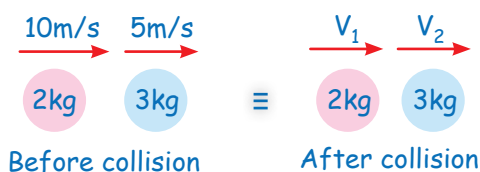


Q. Two ball 2 kg and 3 kg are moving with velocity 10 m/s and 5 m/s as shown in figure. Find velocity of ball after collision if collision is

(a) Elastic ($e = 1$)

(b) Inelastic ($e = 0.5$)

(c) Perfectly inelastic ($e = 0$)



Sol. (a) Elastic ($e = 1$)

$$2 \times 10 + 3 \times 5 = 2V_1 + 3V_2 \quad \dots(i)$$

$$e = 1 = \frac{V_2 - V_1}{5} \quad \dots(ii)$$

Now you can solve both eqn. and get ans.

(b) Inelastic ($e = 0.5$)

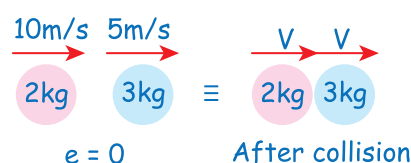
$$2 \times 10 + 3 \times 5 = 2V_1 + 3V_2$$

$$e = \frac{1}{2} = \frac{V_2 - V_1}{5}$$

Now you can solve both eqn. and get ans.

$$\text{loss in (KE)} = (KE)_f - (KE)_i$$

(c) Perfectly inelastic ($e = 0$)



$$P_i = P_f$$

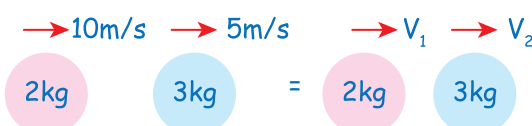
$$2 \times 10 + 3 \times 5 = 2V + 3V$$

$$V = 7$$

$$\text{loss in KE} = (KE)_f - (KE)_i$$

$$= \left[\frac{1}{2} (2+3) 7^2 \right] - \left[\frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 5^2 \right]$$

Q. Two balls are moving with is speed 10 m/s and 5 m/s as shown in figure. Find velocity of each ball after collision if collision is elastic ($e = 1$) also find max. potential energy and mini. kinetic energy during collision.



Bhai, aise ques. mein remember do eqn bahut use hogi: ek momentum conservation ki aur doosri ' e ' ki definition ki.

$$\rightarrow \vec{P}_i = \vec{P}_f$$

$$2 \times 10 + 3 \times 5 = 2v_1 + 3v_2 \quad \dots(i)$$

$$e = 1 = \frac{V_2 - V_1}{5} \quad \dots(ii)$$

Solve (1) and (2)

$$V_1 = 4 \text{ and } V_2 = 9$$

During the collision potential energy will be maximum when velocity of both balls become equal.

PE max tab jab dono ki V same jitni
(KE) kam utni (PE) aa jaayegi.
P conserve kare isse V nikale aur
 ΔKE aa jaayegi



$$P_i = P_f$$

$$2 \times 10 + 3 \times 5 = 2V + 3V$$

$$V = 7$$

$$(KE)_{\text{before}} = \frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 5^2 = \frac{275}{2}$$

$$(KE)_{\text{now}} = (KE)_{\text{min}} = \frac{1}{2} \times 2v^2 + \frac{1}{2} \times 3v^2 = \frac{245}{2}$$

$$\Delta KE = \frac{245}{2} - \frac{275}{2} = -15$$

Hence

$$PE (\text{maximum}) = 15$$

$$(KE)_{\text{min}} = \frac{245}{2}$$

Le bhai ek sawaal aur le and concept bilkul clear ho
jaana chahiye



Q. Two balls of masses 2 kg and 4 kg are moving towards each other with velocities 4 m/s and 2 m/s respectively on a frictionless surface. After colliding the 2 kg balls returns back with velocity 2m/s. Then find:

- velocity of 4 kg ball after collision
- coefficient of restitution e .
- Impulse of deformation J_D .
- Maximum potential energy of deformation.
- Impulse of reformation J_R .

Just before collision

Just after collision



Sol. (a) By momentum conservation,

$$2(4) - 4(2) = 2(-2) + 4(v_2)$$

$$\Rightarrow v_2 = 1 \text{ m/s}$$

$$(b) e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{1 - (-2)}{4 - (-2)} = \frac{3}{6} = 0.5$$

(c) At maximum deformed state, by conservation of momentum, common velocity is $v = 0$.

$$J_D = m_1(v - u_1) = m_2(v - u_2)$$

$$= 2(0 - 4) = -8 \text{ N-s}$$

$$= 4(0 - 2) = -8 \text{ N-s}$$

$$\text{or } = 4(0 - 2) = -8 \text{ N-s}$$

(d) Potential energy at maximum deformed state U = loss in kinetic energy during deformation.

$$\text{or } U = \left(\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \right) - \frac{1}{2} (m_1 + m_2) v^2$$

$$= \left(\frac{1}{2} 2(4)^2 + \frac{1}{2} 4(2)^2 \right) - \frac{1}{2} (2 + 4)(0)^2$$

$$\text{or } U = 24 \text{ Joule}$$

$$(e) J_R = m_1(v_1 - v) = m_2(v - v_2)$$

$$= 2(-2 - 0) = -4 \text{ N-s}$$

$$\text{or } = 4(0 - 1) = -4 \text{ N-s}$$

$$\text{or } e = \frac{J_R}{J_D}$$

$$\Rightarrow J_R = e J_D$$

$$= (0.5)(-8) = -4 \text{ N-s}$$

Q. m collides with $2m$. Coefficient of restitution is 1. Find V_1 & V_2 if m was moving with velocity V before collision.



$$e = 1$$

$$P_i = P_f$$

$$m \times v + 0 = mv_1 + 2mv_2$$

$$V = V_1 + 2V_2 \quad \dots(1)$$

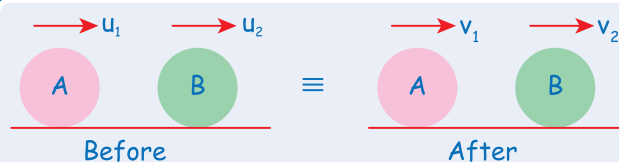
$$e = 1 = \frac{V_2 - V_1}{V} \Rightarrow V_2 - V_1 = V \quad \dots(2)$$

Solve and get

$$v_2 = \frac{2v}{3}, v_1 = \frac{-v}{3}$$



Fractional decrease in kinetic energy of the colliding particle = 8/9 (do yourself)



$$P_{\text{before}} = P_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

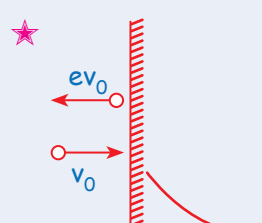
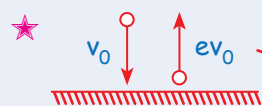
$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

★ If $m_1 = m_2$ and $e = 1$ then after solving we got $v_1 = u_2$ and $v_2 = u_1$

Means if mass is same then in elastic collision ($e = 1$) velocity will exchange



★ If $m_1 \ll m_2$ then $v_1 \approx -eu_1$

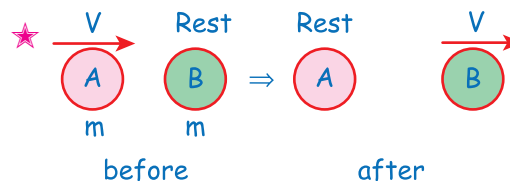
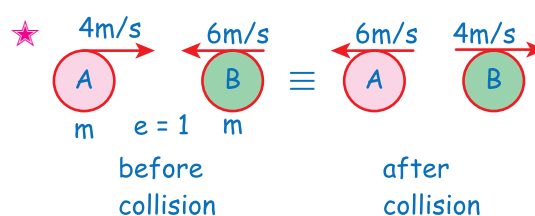


$$|\Delta \vec{P}| = mv_0(1+e)$$

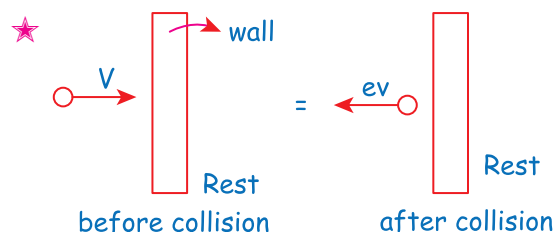
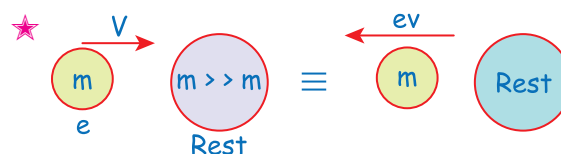
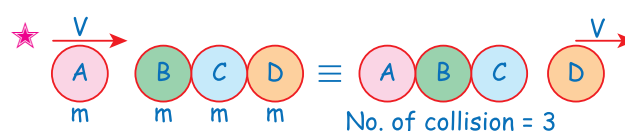
If $e = 1$, $|\Delta \vec{P}| = 2mV_0$



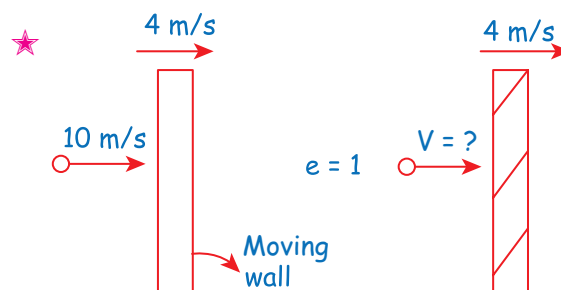
Matlab, head-on elastic collision mein agar mass same hai toh velocity ki adla badli ho jaayegi aur agar koi mass 'v' velocity se kisi bahut bhaari mass se takraayega toh 'ev' se waapas aayega



Q. Find velocity of A, B, C, D, after collision. If $e = 1$ for all collision



Q. ball collides with wall moving at 4m/s $e = 1$, find V.

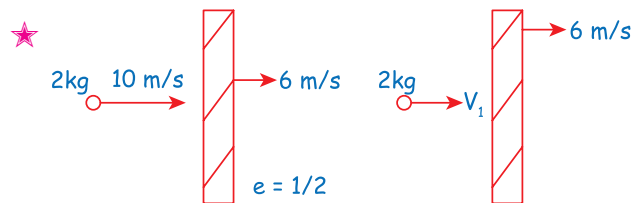


$$e = \frac{4 - V}{10 - 4} \quad \therefore 1 = \frac{4 - V}{6}$$

$$-V = 6 - 4$$

$$V = -2 \rightarrow \text{पीछे}$$

Q. ball collides with wall moving at 6 m/s $e = 1/2$. Find impulse to ball.



$$e = \frac{6 - V_1}{4} = \frac{1}{2}$$

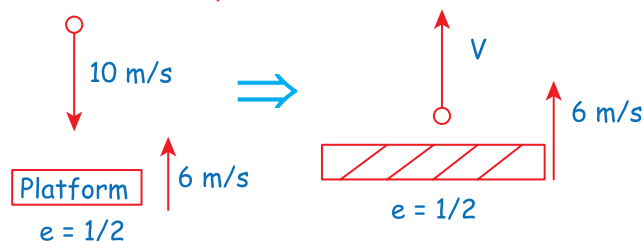
$$V_1 = 4$$

$$(\Delta \vec{P})_{\text{ball}} = (2 \times 4) - (2 \times 10) = -12$$

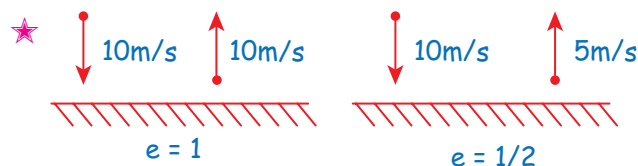
$$\text{Impulse} = -12$$

Wall ko velocity par kuch khaas fark nahi padega bez mass of wall is very large.

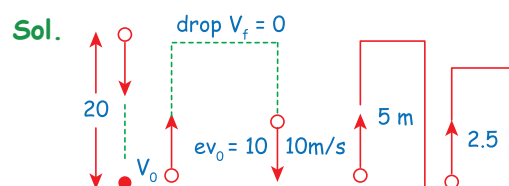
Q. Ball & platform collide as shown in figure. $e = 1/2$. Find velocity of ball after collision.



$$\text{Sol. } e = \frac{1}{2} = \frac{V - 6}{16} \Rightarrow V = 14$$



Q. A ball is dropped from 20m height on a floor, $e = 1/2$. Find velocity of ball just after 3rd collision.



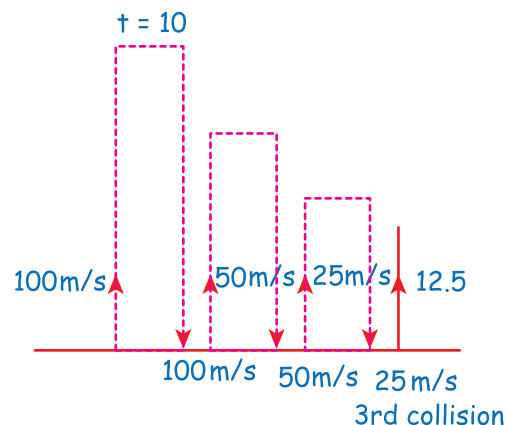
Velocity before hitting floor 1st time

$$V_0^2 = 0^2 + 2 \times 10 \times 20$$

$$V = 20$$

ball jameen se 20 m/s se takraegi aur 10 m/s se rebound hogi bcz $e = 1/2$ so, 2nd collision mein 5 m/s se rebound hogi aur har collision mein uski speed half hoti jaegi so, ans = 2.5.

Q. A ball is thrown up with 100m/s as shown. Coefficient to restitution $e = 1/2$

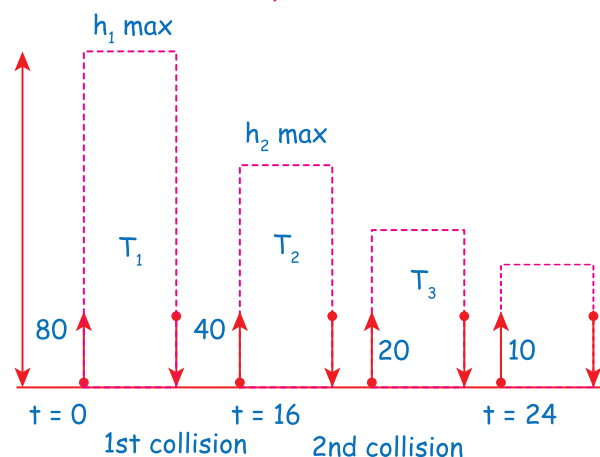


Find when particle will strike floor 3rd time

$$\text{Sol. } (10 + 10) + (5 + 5) + (2.5 + 2.5)$$

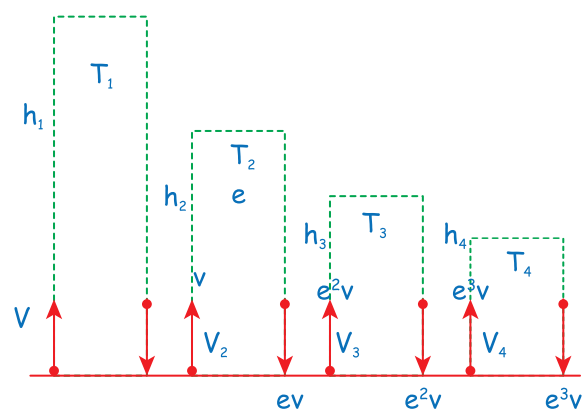
$$t = 26$$

Q. A ball is thrown up with 80 m/s. $e = 1/2$



Q. Ball thrown up with velocity V. Coefficient of restitution = e

Sol.



$$V \Rightarrow 1 : e : e^2 : e^3 = V_1 : V_2 : V_3 : V_4$$

$$T \Rightarrow 1 : e : e^2 : e^3 = T_1 : T_2 : T_3 : T_4$$

$$h_{\max} \Rightarrow 1 : e^2 : e^4 : \dots =$$

$$T = \frac{2V_y}{g} \quad h_{\max} = \frac{V_y^2}{2g}$$

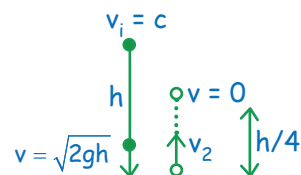
$$(h_1)_{\max} = \frac{v^2}{2g}$$

$$(h_2)_{\max} = \frac{(ev)^2}{2g}$$

$$(h_3)_{\max} = \frac{(e^2v)^2}{2g}$$

- Q.** Two balls of same mass are dropped from the same height h , on to the floor. The first ball bounces to a height $h/4$ after the collision & the second ball to a height $h/16$. The impulse applied by the first & second ball on the floor are 1 and Irrespectively. Then

Sol.

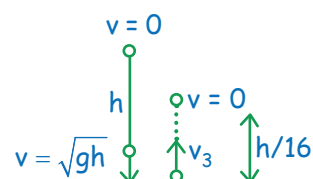


$$0^2 = v_2^2 - 2g \cdot h/4$$

$$\Rightarrow v_2 = \sqrt{gh/2}$$

$$I_1 = \bar{P}_f - \bar{P}_i = m\sqrt{\frac{gh}{2}} + m\sqrt{2gh}$$

$$I_1 = \frac{3m\sqrt{gh}}{\sqrt{2}}$$



$$0^2 = v_3^2 - 2g \frac{h}{16} \Rightarrow v_3 = \frac{\sqrt{gh}}{2\sqrt{2}}$$

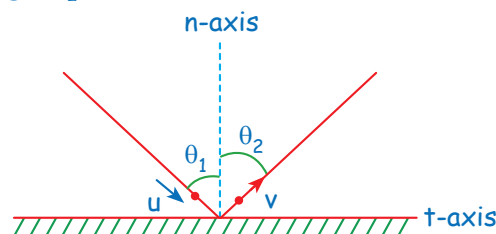
$$\therefore I_2 = m \cdot \frac{\sqrt{2gh}}{4} + m\sqrt{2gh}$$

$$I_2 = \frac{5m\sqrt{2gh}}{4}$$

$$\frac{I_1}{I_2} = \frac{3}{5\sqrt{2}} \Rightarrow 5I_1 = 6I_2$$

Oblique Collision of a Particle with a Fixed Smooth Surface

Consider a ball moving with velocity u at an angle θ_1 with normal to the surface and rebounds with velocity v at angle θ_2 with the normal.



Since the surface is smooth, there is no impulse along the surface. Therefore,

$$u_t = v_t \Rightarrow u \sin \theta_1 = v \sin \theta_2$$

$$v_n = eu_n \Rightarrow eu \cos \theta_2 = v \cos \theta_1$$

$$\Rightarrow \tan \theta_2 = \frac{\tan \theta_1}{e}$$

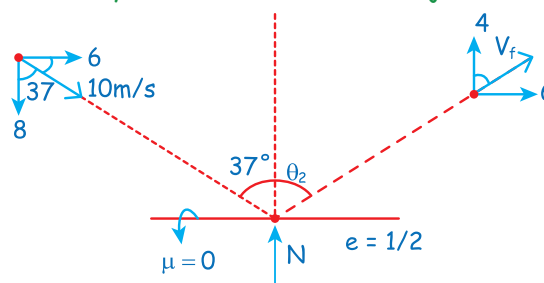
$$v = u \sqrt{\sin^2 \theta_1 + e^2 \cos^2 \theta_1}$$

For elastic collision $e = 1$

$$v = u \text{ and } \theta_2 = \theta_1$$

- Q.** A ball is moving on horizontal plane with velocity 10 m/s strike frictionless surface at an angle 37° with normal to the surface. Find velocity of ball after collision.

Sol. Velocity before collision = $6\hat{i} - 8\hat{j}$



$$\bar{V}_f = 6\hat{i} + 4\hat{j}$$

$$\tan \theta_2 = \frac{6}{4} = \frac{3}{2}$$

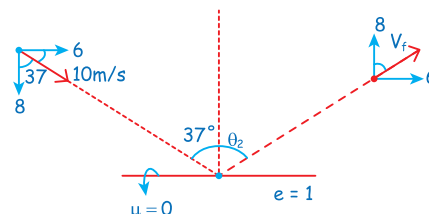
$$(\theta_2 \neq 37^\circ)$$

- Q.** Repeat the above problem if $e = 1$

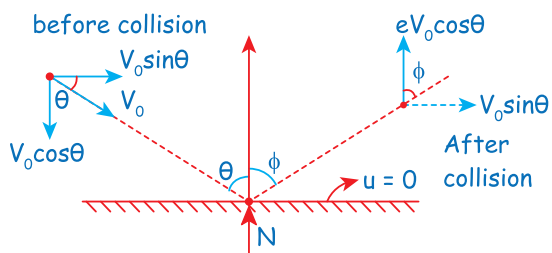
Sol. $v_i = 6\hat{i} - 8\hat{j}$

$$\bar{V}_f = 6\hat{i} + 8\hat{j}$$

$$(\theta_2 = 37^\circ)$$



I hope now you can understand following diagram.



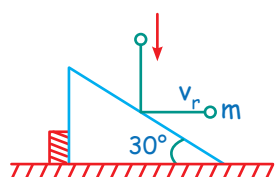
$$\tan \phi = \frac{V_0 \sin \theta}{e V_0 \cos \theta}$$

$$\tan \phi = \frac{\tan \theta}{e}$$

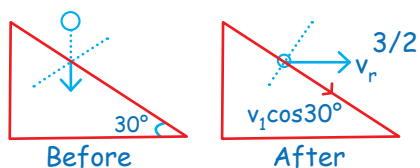
$$\text{If } e = 1 \Rightarrow \phi = \theta$$

$$0 < e < 1 \Rightarrow \phi > \theta$$

- Q.** As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30° , the velocity v_f will be



Sol. $v_f \cos 30^\circ = \frac{3}{2} \Rightarrow v_f = \frac{\sqrt{3}}{2} \text{ m/s}$



- Q.** A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What cannot be the possible angle of its velocity with the vertical after the collision. (Assume $e < 1$).

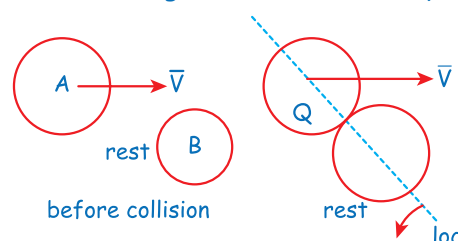


Sol. $\tan \theta = \frac{v_0}{e v_0} = \frac{1}{e} \because e < 1$

$$\tan \theta \geq 1 \Rightarrow \theta \geq 45^\circ$$

Oblique Collision

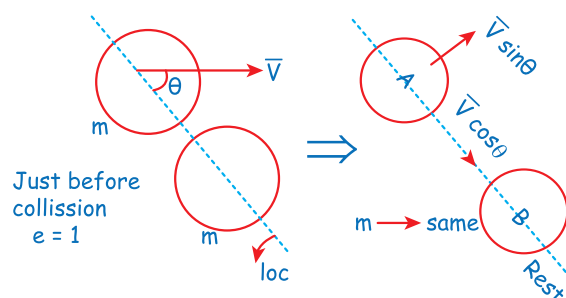
★ When v is not along LOC Collision is oblique collision



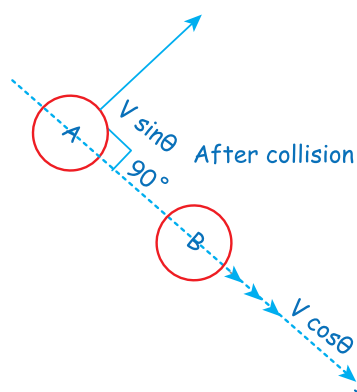
SKC

Oblique collision mein sabse pehle LOC draw karo aur collision se pehle dono body ki velocity ke component LOC ki taraf aur perpendicular todlo. Along the LOC wahi purane wale head-on collision ke concept lagado aur perpendicular wali velocity as it is rahne do. Bas itna hi hai, ab bolo... ye tough hai ya aasaan

- Q.** A ball of mass m moving with velocity V collides with another ball of mass m at rest elastically and obliquely. Find angle between their direction of motion finally.

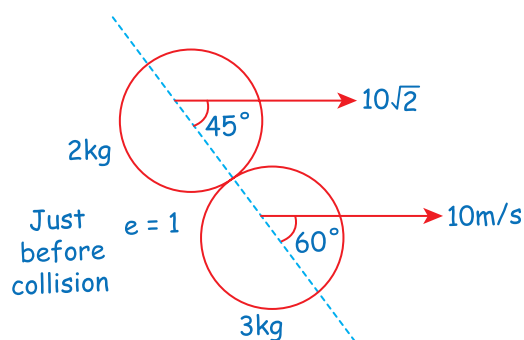


Since, mass is same and collision is elastic so, along the LOC velocity will interchange

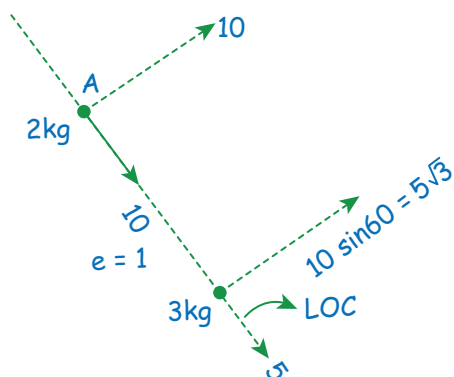


both move in $\perp^r$ directn to each other

Q. Two bodies are colliding as shown in figure. Find final velocities after collision.



Sol.



Momentum conservation along LOC

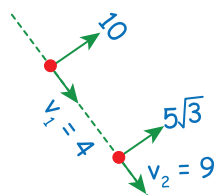
$$P_i = P_f$$

$$20 + 15 = 2v_1 + 3v_2$$

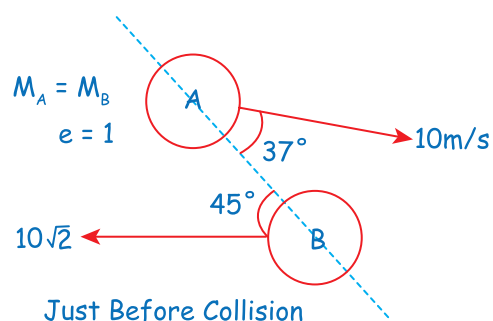
$$e = 1 = \frac{V_2 - V_1}{5}$$

Solve and get

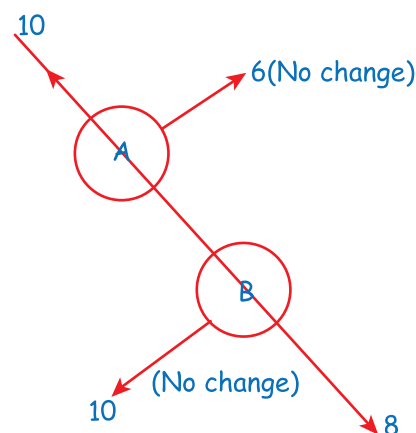
$$V_1 = 4 \text{ \& } V_2 = 9 \text{ so,}$$



Q. Two bodies are colliding as shown in figure. Find velocity of A and B after collision.



Sol.



Impulsive force → Those force which can change the momentum of body by significant amount in very small interval of time

Non-Impulsive force → Those forces which bring very small change in momentum for very small interval of time are called non-Impulsive force.

ex → mg (gravitational force), spring force

Hence we can say

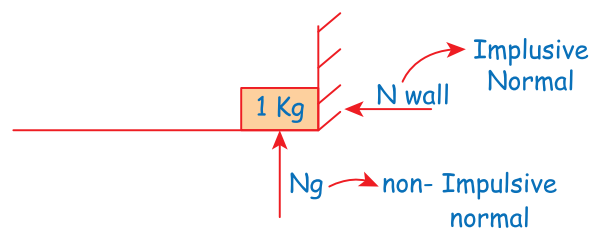
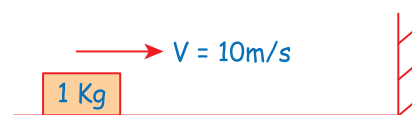
$\vec{P} \rightarrow$ conserve

(1) When net ext force = 0

(2) When non-impulsive force is considered for very small interval of time, momentum will approx. conserve.

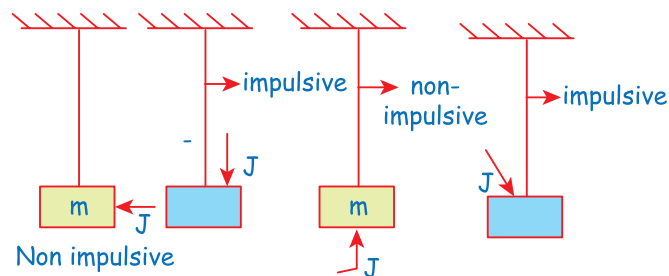
Normal Force

★ Collision ki traf wala normal always impulsive

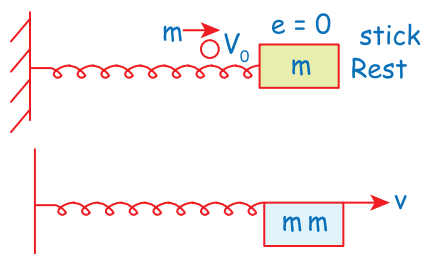


★ If normal is impulsive then corresponding friction will also impulsive $f = \mu N$

Tension



Q. A particle of mass m moving with velocity V collides with block at rest & stick with it. Find their combined velocity.

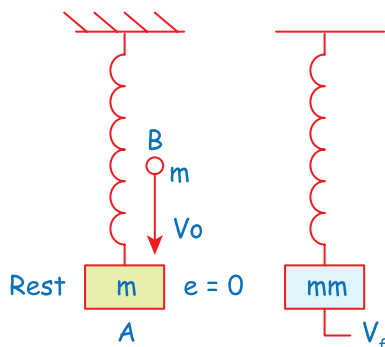


(P_i) just before = (P_f) just after

$$mV_0 + 0 = 2mV_f$$

$$V_f = V_0/2$$

Q. Particle of mass m collides with block as shown. Find velocity after after collision.



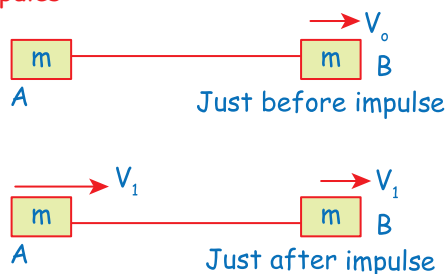
$(A + B) \rightarrow$ system

(P_i) Just before = (P_f) just after collision

$$mV_0 + 0 = 2mV_f$$

$$V_f = V_0/2$$

Q. Two blocks are connected by a massless string. Block B is given impulse. Find velocity after impulses



$$P_i = P_f$$

$$0 + mV_0 = mV' + mV'$$

$$V' = \frac{V_0}{2}$$

Ab hum displacement of COM par based ques dekhenge jo ki almost same pattern par hote hain. Concept yah hai jis disha mein net external force 0 hai aur COM initially rest par hai to COM hamesha rest par rahega and displacement of COM will be 0

If initially com is at rest and $(F_{net})_{ext} = 0$ then

$(\bar{d}_{com})$ displacement of COM = 0

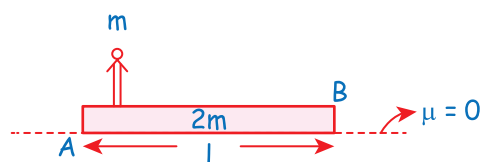
$$\bar{d}_{com} = \frac{m_1 \bar{d}_1 + m_2 \bar{d}_2}{m_1 + m_2} + \dots = 0$$

$$\Rightarrow m_1 \bar{d}_1 + m_2 \bar{d}_2 = 0$$

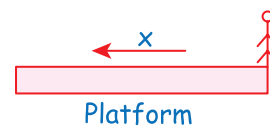


Q. Initially man and platform both are at rest now man start walking and reaches at B.

Find displacement of platform. Ground is smooth



$$\text{Sol. } m_1 \bar{d}_1 + m \bar{d}_2 = 0$$



Let platform piche x gya

$$-2mx + m(l - x) = 0$$

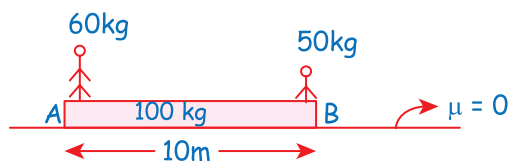
$$L = 3x, \quad x = \frac{L}{3}$$

$$(\bar{d}_{com})_{\text{man wrt ground}} = \frac{2L}{3}$$

Agar main platform or man ko system maanu to ext. force is zero. Friction to hai but internal hai aur initially COM rest par hai. So, displacement of COM = 0



Q. Initially all are at rest. Both come closer and they reach at center of platform. Find displacement of Platform.



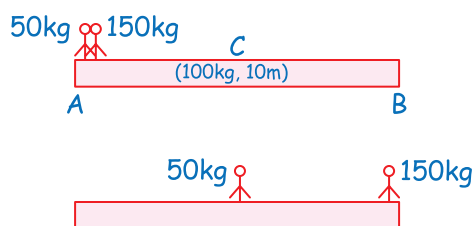
Sol. Man lo Platform piche x gya

$$m_1 \vec{d}_1 + m_2 \vec{d}_2 + m_3 \vec{d}_3 = 0$$

$$-100x + 60(5 - x) - 50(x + 5) = 0$$

$$x = \frac{5}{21}$$

Q. Initially all are at Rest. Both person are on platform such that 50 kg boy reaches at centre of platform C and 150 kg boy reaches at point B. Find displacement of com of platform



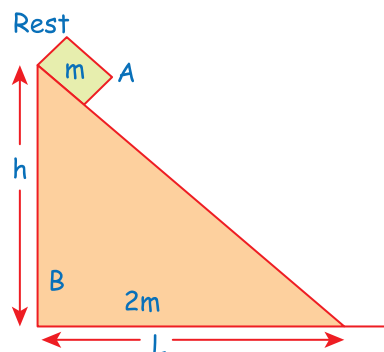
Sol. Man lo Platform piche x gya

$$-100x + 50(5 - x) + 150(10 - x) = 0$$

$$2x + (x + 5) + 3x + 30$$

$$6x = -35 = x = \frac{-35}{6}$$

Q. A block of mass m is placed on wedge B and released as shown in figure. If all surface are smooth and wedge is free to move find displacement of wedge when block reaches at ground.



Yah bahut important ques hai jo baar-baar use hoga

Sol. $(\vec{F}_{\text{net}})_x = 0$

Net force along horizontal is zero and initially COM of system is at rest so displacement of COM will be zero along x-axis.

$$m_1 \vec{d}_1 + m_2 \vec{d}_2 = 0$$

Man lo wedge piche x gya

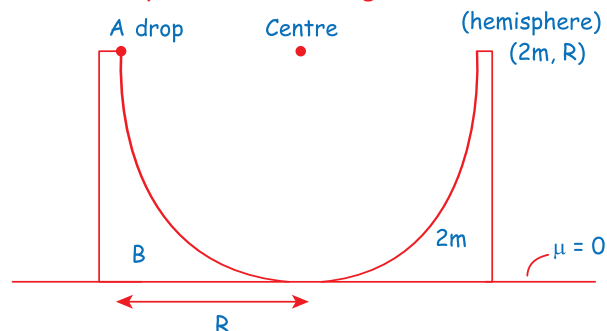
$$(-2mx + m(L - x) = 0) \quad [\text{along horizontal}]$$

$$x = \frac{L}{3}$$

SKC

We carefull overall yahan displacement of COM zero nahi hai Bcz jab block niche aayega to system ka COM bhi niche aayega but x-axis me displacement of mass zero hai so, we can say centre of mass system is moving downward in a straight line path till block reaches at ground

Q. When ball of mass x reaches at lowest point, find displacement of wedge B.



Sol. $m_1 \vec{d}_1 + m_2 \vec{d}_2 = 0$

Man lo wedge piche x gya

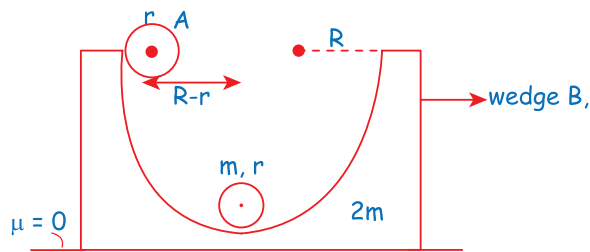
$$(-2mx + m(R - x) = 0) \quad [\text{along horizontal}]$$

$$x = \frac{R}{3}$$

Agar hum dhyani se dekhe to platform par man walking, upar ke do ques. aur niche wala sawal bhi same pattern par hai bas horizontal distance alag alag hai. Bas ye dhyani rakhna hum displacement nikal rahe hai na ki velocity.



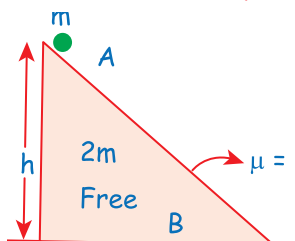
Q. When sphere A reaches at lowest point find displacement of B.



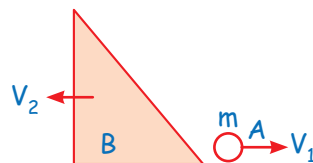
Sol. $-2mx + m(R - r - x) = 0$

$$x = +\frac{(R-r)}{3} \quad (F_{net})_{ext} 'X' = 0$$

Q. A ball of mass m is released from point A from a height h on the inclined plane as shown in figure. Find velocity of ball and wedge when ball reaches at lowest point V_1 & V_2 at lowest point



Sol.



$$P_i = P_f \text{ (x में)}$$

$$0 + 0 = mv_1 - 2mv_2$$

$$v_1 = 2v_2$$

...(1)



$$(WET) W_g + W_N + W_f = \Delta KE$$

$$(mgh + 0) + 0 + 0 = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2 \quad \dots(2)$$

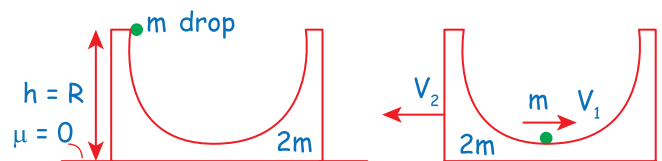
Solve and get V_1 & V_2

Net (WD) by common normal = 0

$$(WD) \text{ by common normal on wedge} = \frac{1}{2} \cdot 2m \cdot v_2^2$$

$$(WD) \text{ on ball by normal} = \frac{-1}{2} \times 2mv_2^2$$

Q. A small ball is released as shown in figure. Find velocity of ball and wedge when it reaches lowest point.



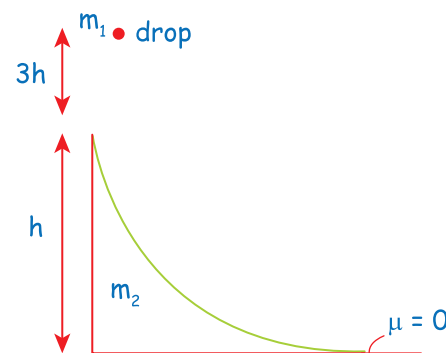
$$mv_1 - 2mv_2 = 0 \quad \dots(1)$$

$$+mgR + 0 + 0 = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2 \quad \dots(2)$$

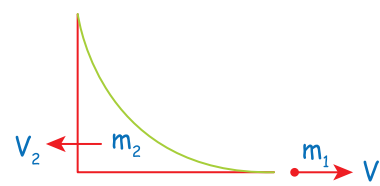
When ball reaches the lowest point N_{ground} will be greater than $3mg$ (socho ye kaise hua)



Q. Find velocity of m_1 when it reaches ground.



Sol.



$$P_i = P_f \text{ (x में)}$$

$$0 = m_1v_1 - m_2v_2 \quad \dots(1)$$

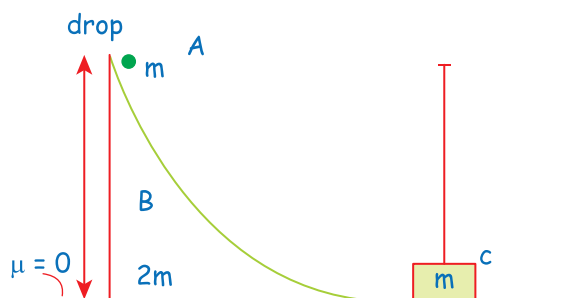
$$m_1v_1 = m_2v_2$$

$$W_g + W_N + W_f = \Delta KE$$

$$\left(m_1g_1 4h + 0 + 0 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \right) \quad \dots(2)$$

Now you can solve i hope you got concept.

Q. A ball A is released from an wedge which is free to move. When ball reaches the ground it is strike elastically the block C of same mass as that of ball. Find the minimum value of h so that block C completes vertical circular motion.



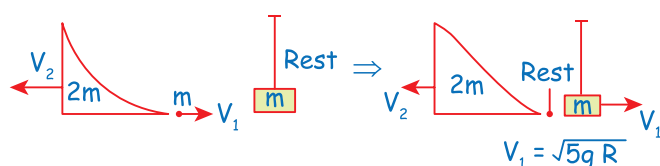
Dekh bhai momentum conservation aur WET (kaleen bhaiya) laga kar A aur B ki velocity nikal lega na.... abe ye to kitni baar kar chuke hai ab bas A aur C ka collision krana hai since, $m_A = m_C$ so, velocity exchange toh $v_1 = \sqrt{5gR}$ hi karna hai kya tum kr skte ho

chlo krke dikhao if $m = 2 \text{ kg}$, $R = 2 \text{ m}$

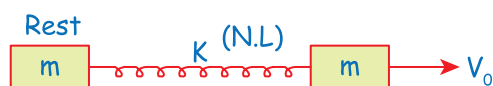
Find h minimum

Ans. $h = 7.5 \text{ m}$

are tum to kr loge na



Q. For the blocks shown as in figure spring is in natural length



Find max elongation in spring.



At max elongation velocities are equal

initial $\bar{P}_i = P_f$ (final momentum)

$$0 + mv_0 = mv + mv$$

$$v = \frac{v_0}{2}$$

$$W_g + W_N + W_{sp} = \Delta KE$$

$$0 + 0 - \frac{1}{2}k(x_{\max}^2 - 0^2) = \left[\frac{1}{2}m\left(\frac{v_0}{2}\right)^2 \times 2 - \left(\frac{1}{2}mv_0^2\right) \right]$$

$$-\frac{1}{2}kx_{\max}^2 = \frac{-mv_0^2}{4}$$

$$x_{\max} = v_0 \sqrt{\frac{m}{2k}}$$

Q. Find max compression in following figure.

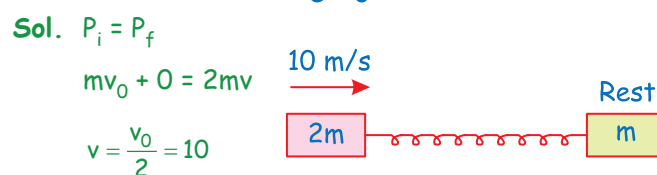
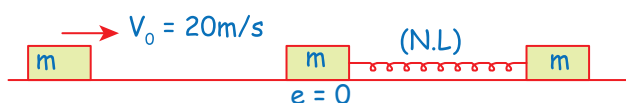


$$P_i = P_f$$

$$0 + mv_0 = mv + mv_0$$

$$x_{\max} = v_0 \sqrt{\frac{m}{2k}}$$

Q. Block with velocity 20m/s collides inelastically with another block find velocity of block just after collision and maximum compression in spring.



Now you can solve it for to calculate maximum compression or elongation... Are ye upar jaisa hi to sawal ho gaya bas do eqn banao ek momentum conservation aur doosri WET solve and get



Q. Find max elongation blocks move as shown in fig.



Sol. Elongation will be maximum when velocity of both the block will be same



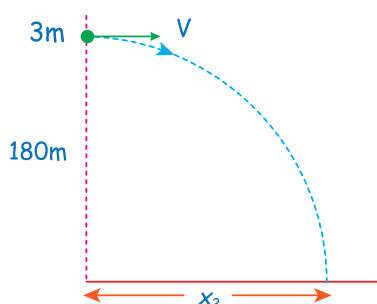
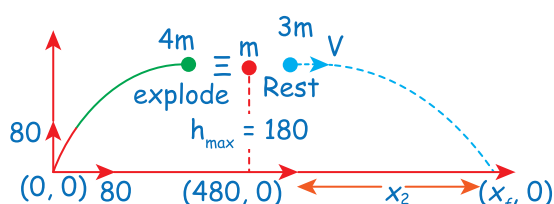
Momentum conservation

(1) $(P_i)_{\text{just before collide}} = (P_f)_{\text{after collide}}$

$$4m \times 80\hat{i} = m \times 0 + 3mV_f$$

$$V_f = \frac{320}{3}$$

$$h_{\max} = \frac{60^2}{2g} = 180$$



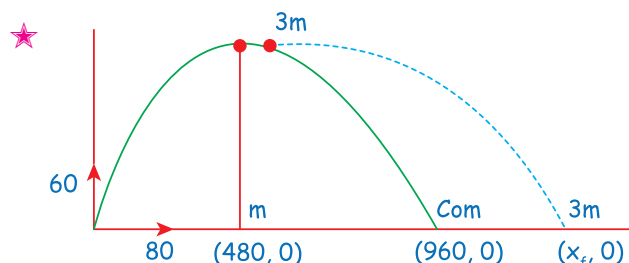
$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 180}{10}} = 6$$

$$x_2 = Vt = \frac{320 \times 6}{3}$$

Co-ordinate of landing point of $3m = 480 + x_2$

$$= 480 + 640 = 1120$$

$$M-II \quad x_2 = 640$$



$$x_{\text{com}} = \frac{mx_1 + m_2x_2}{m_1 + m_2}$$

$$960 = \frac{m_1 \times 480 + 3m \times x_f}{4m}$$

$$x_f = \frac{3360}{3} = 1120$$

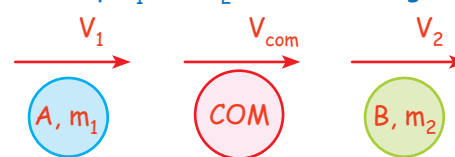
(m, 3m $\Rightarrow$ dono ek sath jameen par takraye)

COM FRAME



COM frame ek phodu article hai jiske saath agar aap khelna sikh gaye to COM ka maza dugna ho jata hai. Kuch question hum COM frame se bahut kam step mein solve kar lete hain, lekin exam ke POV se aap isse last priority mein rakhe it's for Adv only (NEET, Mains wale isse skip kar sakte hain)

★ Suppose two ball of mass m_1 and m_2 are moving with velocity V_1 and V_2 as shown in fig.



$$\vec{V}_{\text{com}} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$$\begin{aligned} (\vec{p}_{1/\text{com}}) &\rightarrow m_1\vec{v}_{1/\text{com}} = m_1 \left[\vec{v}_1 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \right] \\ &= m_1 \left[\frac{m_1\vec{v}_1 + m_2\vec{v}_2 - m_1\vec{v}_1 - m_2\vec{v}_2}{m_1 + m_2} \right] \\ &= \frac{m_1m_2(\vec{v}_1 - \vec{v}_2)}{(m_1 + m_2)} \end{aligned}$$

$$(\vec{p}_{2/\text{com}}) = m_2\vec{v}_{2/\text{com}} = m_2 \left[\vec{v}_2 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \right]$$

$$\frac{m_1m_2(\vec{v}_2 - \vec{v}_1)}{m_1 + m_2}$$

$$\vec{v}_{1/\text{com}} = \vec{v}_1 - \vec{v}_{\text{com}} = \frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1 + m_2}$$

$$\vec{p}_{1/\text{com}} + \vec{p}_{2/\text{com}} = 0$$

Two Particles System

$$(KE)_{1/\text{com}} = \frac{1}{2}m_1(\vec{v}_{1/\text{com}})^2 = \frac{1}{2}m_1 \left(\frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1 + m_2} \right)^2$$

$$(KE)_{2/\text{com}} = \frac{1}{2}m_2(\vec{v}_{2/\text{com}})^2 = \frac{1}{2}m_2 \left(\frac{m_1(\vec{v}_2 - \vec{v}_1)}{m_1 + m_2} \right)^2$$

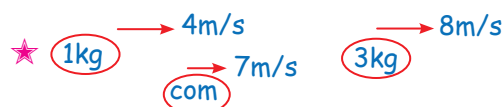
Total KE of system Incom frame

$$= \frac{1}{2} \cdot \frac{m_1 m_2^2 (v_1 - v_2)^2}{(m_1 + m_2)^2} + \frac{1}{2} \cdot \frac{m_2 m_1^2 (v_2 - v_1)^2}{(m_1 + m_2)^2}$$

$$= \frac{1}{2} \frac{(m_1 m_2)}{(m_1 + m_2)^2} (v_1 - v_2)^2 (m_2 + m_1)$$

$$= \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (v_1 - v_2)^2$$

$\mu = \text{reduced mass}$



$$V_{\text{com}} = \frac{1 \times 4 + 3 \times 8}{1 + 3} = \frac{4 + 24}{4} = 7$$

$$\vec{V}_{A/C} = \vec{V}_{A/\text{com}} = (4 - 7)\hat{i} = -3\hat{i}$$

$$(\vec{P}_A)_{\text{incomframe}} = 1 \times (-3\hat{i}) = -3\hat{i}$$

$$\vec{V}_{B/C} = (8 - 7)\hat{i} = \hat{i}$$

$$(\vec{P}_B)_{\text{incomframe}} = 3 \times 1\hat{i} = +3\hat{i}$$

momentum of system

$$\text{In com frame} = (-3\hat{i}) + (3\hat{i}) = 0$$

In com frame total momentum of system will be always zero

Total K.E of system in ground frame

$$\frac{1}{2} \times 1 \times 4^2 + \frac{1}{2} \times 3 \times 8^2 = 104$$

Total K.E of system in com frame

$$\frac{1}{2} \times 1 \times (3)^2 + \frac{1}{2} \times 3 \times (1)^2 = 6$$

$$\text{Total K.E of COM} = \frac{1}{2} \times 4 \times (7)^2 = 98$$



Kaam ka Dabba

In com frame Total KE

$$\Rightarrow \frac{1}{2} \mu v_{\text{relative}}^2$$

$$\left(\mu = \frac{m_1 m_2}{m_1 + m_2} \equiv \text{Reduced mass} \right)$$

ye formula
apko nhi
bholna hai



For two Particle system

① In com frame total KE of system

$$= \frac{1}{2} \mu v_{\text{relative}}^2 \quad \left(\mu = \frac{m_1 m_2}{m_1 + m_2} \right)$$

② Total KE of system = (KE) of system + (KE)_{com}

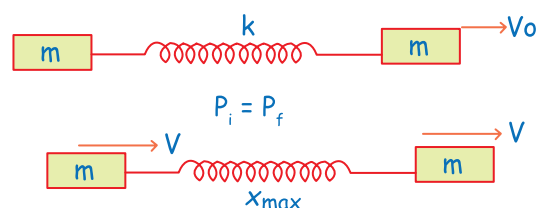
In com frame

$$= \frac{1}{2} \mu v_{\text{rel}}^2 + \frac{1}{2} m_{\text{total}} v_{\text{com}}^2$$

$$(KE)_{\text{system}} = \frac{1}{2} \mu v_{\text{rel}}^2 + \frac{p^2}{2m}$$

$$\text{If } (\vec{F}_{\text{net}})_{\text{ext}} = 0 \Rightarrow \Delta(K.E) = \frac{1}{2} \mu \left[(v_{\text{rel}})_f^2 - (v_{\text{rel}})_i^2 \right]$$

Q. Find max elongation in spring.



Sol.

Method-I

$$0 + mv_0 = mv + mv \Rightarrow v = \frac{v_0}{2}$$

WET

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \cdot m \left(\frac{v_0}{2} \right)^2 \times 2 - \frac{1}{2} m v_0^2$$

$$x_{\text{max}} = v_0 \sqrt{\frac{m}{2k}}$$

Method-II (In COM frame)

$$0 + mv_0 = mv + mv \Rightarrow v = \frac{v_0}{2}$$

WET (work energy theorem in COM Frame)

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \mu \left[(v_{\text{rel}})_f^2 - (v_{\text{rel}})_i^2 \right]$$

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \frac{m}{2} \left[(0)^2 - (v_0)^2 \right]$$

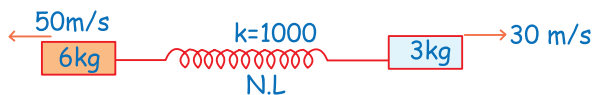
$$x_{\text{max}} = v_0 \sqrt{\frac{m}{2k}}$$

$$\mu = \frac{m \cdot m}{m + m} = \frac{m}{2}$$

Ho sakta hai kuch baccho ko COM frame wala method tough lage, believe me agar acche se practice ki to this method will save your time jaise agle ques mein we already know elongation max. tab hoga jab dono block ki velocity same ho jyegi means $(V_{rel})_f = 0$. So, direct COM frame me WET me value hi to put krni hai jaise.....



Q. Find max elongation = ?



$$\mu = \frac{6 \times 3}{6 + 3} = 2$$

$$-\frac{1}{2} \times 1000 \times (x_f^2 - 0^2) = \frac{1}{2} \times 2 \times (0^2 - (80)^2)$$

Variable Mass System

Suppose dm is added to m .

$$dm \cdot v + m \cdot u = (m + dm) \cdot (u + du)$$

momentum conservation

$$dm \cdot v + m \cdot u = (m + dm) \cdot (u + du)$$

$$vdm + \cancel{m \cdot u} = \cancel{m \cdot u} + mdu + udm + \cancel{dm \cdot du} \quad \text{Neglect}$$

$$vdm = mdu + udm$$

$$vdm - udm = mdu$$

$$(v - u)dm = mdu$$

$$v_{rel} \frac{dm}{dt} = m \frac{du}{dt}$$

$$\left(\frac{dm}{dt} \right) v_{rel} = ma$$

$$v_{rel} \cdot \frac{dm}{dt} = \text{Force}_{\text{thrust}}$$

Rocket Propulsion

dm is ejected with velocity v_{rel} w.r.t rocket

$$\mu = \frac{m + m_2}{m_1 + m_2} = \frac{m}{2}$$

$$m \cdot u = (m - dm) \cdot (u + du)$$

momentum conservation

$$P_i = P_f$$

$$mu = dm \cdot V_g + (m - dm)(u + du)$$

[V_g is velocity of dm w.r.t ground]

$$mu = dm \cdot (v_{rel} + u + du) + mu + (mdu - udm - dmdu)$$

$$v_{rel} = v_{\text{gas rocket}}$$

$$v_{rel} = \vec{v}_{\text{gas}} - \vec{v}_{\text{rocket}}$$

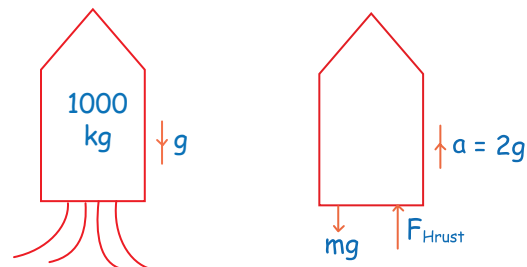
$$mu = v_{rel} \cdot dm + \cancel{u \cdot dm} + \cancel{dm \cdot du} + \cancel{m \cdot du} - \cancel{u \cdot dm} - \cancel{dm}$$

$$-v_{rel} dm = mdu$$

$$-v_{rel} \frac{dm}{dt} = m \frac{dv}{dt} = ma = F_{\text{thrust}}$$

$$F_{\text{thrust}} = -v_{rel} \frac{dm}{dt}$$

★ Gases are ejecting at the rate of 1 kg/sec Find v_{rel} for gas to have an upward acceleration of $2g$ (uplift)



$$F_{\text{thrust}} - mg = ma$$

$$v_{rel} \frac{dm}{dt} - mg = m(2g)$$

$$v_{rel} = 3mg$$

$$v_{rel} = 3 \times 10000 \times 10 = 30000$$

$$m \cdot u = (m - dm) \cdot (u + du)$$

Suppose gas is ejecting at the rate of μ kg/sec

$$\frac{dm}{dt} = \mu = \text{constant}$$

$$dm = \mu dt$$

$$\text{mass of rocket at time 't'} = M = m_0 - \mu t$$

$$m_0 \text{ is the mass of system at } t = 0$$

$$-v_{rel} dm = mdu$$

$$-v_{rel} \mu dt = (m_0 - \mu t) du$$

$$-\int \mu v_{rel} \frac{dt}{m_0 - \mu t} = \int du$$

$$\mu v_{rel} \int_0^t \frac{dt}{m_0 - \mu t} = \int_{u_i}^{v_f} du$$

$$\frac{-\mu v_{rel}}{-\mu} \ln(m_0 - \mu t) \Big|_0^t = v_f - v_i$$

$$v_{rel} [\ln(m_0 - \mu t) - \ln m_0] = v_f - v_i$$

Formula

$$v_f - v_i = v_{rel} \cdot \ln \left(\frac{m_0 - \mu t}{m_0} \right)$$

$$v_f = v_i + v_{rel} \ln \left(\frac{m_0 - \mu t}{m_0} \right)$$

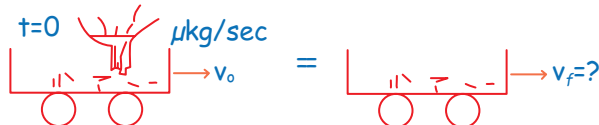
OR

$$= v_i + \ln \left(\frac{m_f}{m_i} \right) \times v_{rel}$$

If $\mu_i = 0$

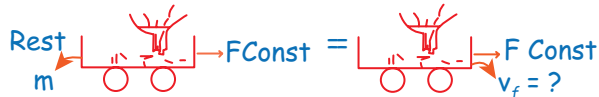
$$v_f = v_{rel} \cdot \ln \left(\frac{m_f}{m_i} \right)$$

Q.



Sol. $mv_0 = (m + \mu t) v_f$ $v_f = \frac{mv}{m + \mu t}$ $t \uparrow, v \downarrow$

Q.



Sol. Impulse = $\Delta P = P_f - P_i$

$$Ft = (m + \mu t) u_f - 0$$

$$v_f = \frac{Ft}{m + \mu t}$$

Ab main advance ke POV se aur aapke concept ko aur strong karne ke liye kuch selected question with proper solution and language ke saath attach kar raha hoon, plz solve them sincerely



Q. A hopper releases grain at a rate dm/dt onto a conveyor belt that moves at a constant speed v . What is the power of the motor driving the belt?

Sol.



Let the system has some arbitrary length of belt whose mass we can call M . This mass of the system increases at the same rate as that the grain falls, so $dM/dt = dm/dt$.

Since the grain falls vertically, $\vec{u} = 0$ and $\vec{v}_{rel} = \vec{u} - \vec{v} = -\vec{v}$, since the speed is constant,

$dv/dt = 0$. Thus from equation for the system of variable mass

$$\vec{F}_{ext} - \vec{v} \frac{dm}{dt} = 0 \Rightarrow \vec{F}_{ext} = \vec{v} \frac{dm}{dt}$$

where $\vec{F}_{ext}$ is the force needed to maintain constant speed because the mass is increasing.

The power required ($P = \vec{F} \cdot \vec{v}$) is

$$P = v^2 \frac{dm}{dt}$$

It is interesting to compare this with the rate at which the kinetic energy of the grain increases.

$$\frac{dK}{dt} = \frac{d}{dt} \left(\frac{1}{2} mv^2 \right) = \frac{1}{2} v^2 \frac{dm}{dt}$$

This is only half the power input. The other half is dissipated as heat when the grain lands on the belt and slips relative to it.

Q. A uniform chain of mass m and length l hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the table on the chain, when half of its length has fallen on the table. The fallen part does not form heap.

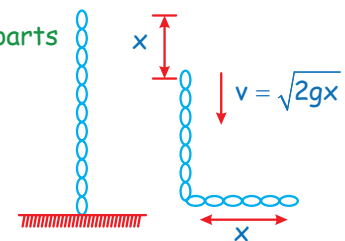
Sol. Let N be the normal reaction acting on the chain.

It consists of two parts

$$N = N_{wt} + N_{thrust}$$

$$N_{wt} = \frac{m}{l} xg$$

$$N_{thrust} = v_{rel} \frac{dm}{dt}$$



If v be the instantaneous velocity of the particle falling on the floor, then

$$v_{rel} = v - 0 = v$$

$$\frac{dm}{dt} = \frac{dm}{dx} \frac{dx}{dt} \text{ or } \frac{dm}{dt} = v \left(\frac{m}{l} \right)$$

$$\text{Thus } N_{th} = v^2 \left(\frac{m}{l} \right), \quad \text{Since } v = \sqrt{2gx}$$

$$\therefore N_{th} = \frac{2mg}{l} x$$

Total reaction is

$$N = \frac{mgx}{l} + \frac{2mgx}{l}, N = 3mg\left(\frac{x}{l}\right)$$

{ $\therefore$ for $x = l/2$ }

$$N = \frac{3}{2}mg$$

- Q.** Two blocks each of mass m , connected by a weightless spring of stiffness k and length L_0 lie on a horizontal surface. A constant force F starts acting on one of the bars (see fig.). Find the maximum distance between the bars during the subsequent motion of the system.



Sol. Consider the system of two blocks and the spring. Force F is external to the system. Therefore neither mechanical energy nor linear momentum will be conserved. Distance between the two blocks will be maximum and minimum when they have the same velocity. Let it be v .

Let d be the distance moved by the right block and x_0 be elongation of the spring when the separation between the blocks is maximum. Consequently, distance moved by the center of mass will be $\left(d - \frac{x_0}{2}\right)$.

The acceleration of the center of mass is given by

$$a_{cm} = \frac{F}{2m}$$

From eqns of constant acceleration, we have

$$v^2 = 2\left(\frac{F}{2m}\right)\left(d - \frac{x_0}{2}\right) \quad \dots (i)$$

and from work-energy theorem, we have

$$\Delta KE + \Delta PE = \text{work done by } F$$

$$\Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}kx_0^2 = Fd \quad \dots (ii)$$

Solving eqn (i) and eqn (ii), we find $x_0 = F/k$

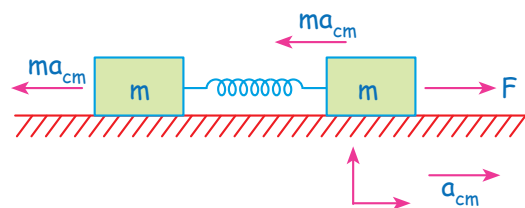
So, the maximum separation is

$$\left(L_0 + \frac{F}{k}\right)$$

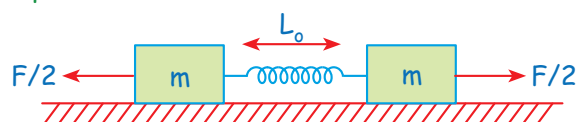
Alternate method(Center of mass frame)

Solving problem in the center of mass frame is always handy as the system is symmetrically placed about the center of mass.

However, a frame fixed to the center of mass in this problem is a non-inertial frame as the acceleration of center of mass is $F/2m$ towards right. Therefore, pseudo forces will act on the blocks towards left.



Setting $a_{cm} = \frac{F}{2m}$, we find that the problem is equivalent to



As the center of mass is at rest in this frame, the blocks move in the opposite direction and come to instantaneous rest at the same time. The extension of the spring is maximum at this instant. Let the two blocks move through distance x_1 and x_2 . The work done on the system by the forces is, therefore

$$W = (F/2)x_1 + (F/2)x_2 = (F/2)(x_1 + x_2)$$

This increases the potential energy of the system. Hence

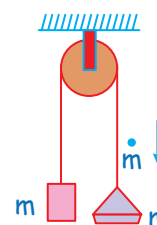
$$(F/2)(x_1 + x_2) = \frac{1}{2}k(x_1 + x_2)^2$$

$$(x_1 + x_2) = F/k$$

The maximum distance between the blocks is, therefore

$$L_0 + x_1 + x_2 = L_0 + F/k$$

- Q.** A block of mass m and a pan of same mass are connected by a string going over a smooth light pulley as shown in figure. Initially the system is at rest until a particle of mass m falls on the pan and sticks to it. If the particle strikes the pan with a speed v find the speed with which the system moves just after the collision.



Sol. Mass m strikes the pan with speed v and stick to it. Let speed of system just after collision is v_1 . For pan and mass m .

Impulse = change in momentum

$$-\int T dt = 2mv_1 - mv$$

For block of mass m

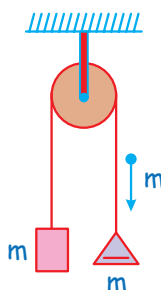
$$\text{impulse} = mv_1$$

$$\int T dt = mv_1$$

$$-mv = 2mv_1 - mv$$

$$mv = 3mv_1$$

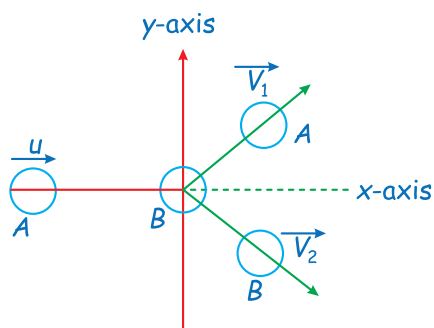
$$v_1 = v / 3$$



Q. A ball A moving with a speed of 9 m/s strikes identical stationary ball B such that after the collision, the direction of each ball makes an angle of 30° with the original line of motion.

(a) Find the speed of the balls after the collision

(b) Is kinetic energy conserved in the collision?



Sol. Choose x-axis along the direction of $\vec{u}$ and y-axis perpendicular to its direction.

Applying conservation of linear momentum along X and Y-axis, we have

$$mu = mv_1 \cos 30^\circ + mv_2 \cos 30^\circ \quad \dots (i)$$

$$\text{and } 0 = mv_1 \sin 30^\circ - mv_2 \sin 30^\circ \quad \dots (ii)$$

After solving eqn (i) and eqn (ii) we find

$$v_1 = v_2 = 3\sqrt{3} \text{ m/s.}$$

(b) Let K_1 and K_2 be the kinetic energy before and after the collision. Then

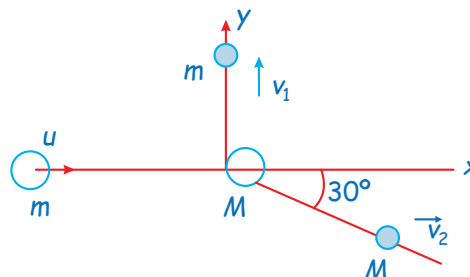
$$K_1 = \frac{1}{2}mu^2 + 0 = \frac{81}{2}m$$

$$\text{and } K_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m(3\sqrt{3})^2 \times 2 = 27m$$

Since $K_2 < K_1$, kinetic energy of the system is not conserved

Q. A particle of mass m having collided with a stationary particle of mass M deviated by an angle $\pi/2$ whereas particle M recoiled at an angle $\theta = 30^\circ$ to the direction of initial motion of particle m . How much and in what way has the kinetic energy changed after the collision if $\frac{M}{m} = 5$?

Sol.



From conservation of momentum in X and Y direction, we have

$$mu = Mv_2 \cos 30^\circ \quad \dots (i)$$

$$0 = mv_1 - Mv_2 \sin 30^\circ \quad \dots (ii)$$

Solving eqn (i) and eqn (ii) for v_1 and v_2 , we get

$$v_1 = u / \sqrt{3}, v_2 = \frac{m}{M} \frac{2u}{\sqrt{3}}$$

Let K_1 and K_2 be the initial and final kinetic energy of the particles. Then

$$K_1 = \frac{1}{2}mu^2$$

$$\text{and } K_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$$= \frac{1}{2}m \left[\frac{1}{3} + \frac{4m}{3M} \right] u^2$$

The percentage loss in K.E. is

$$\frac{K_1 - K_2}{K_1} \times 100 = \frac{2}{5} \times 100 = 40\%$$

Q. A gas molecule having a speed of 300 m/s collides elastically with another molecule of the same mass which is initially at rest. After the collision the first molecule moves at an angle of 30° to its initial direction. Find the speed of each molecule after collision and the angle made with the incident direction by the recoiling target molecule.

Sol. This application corresponds exactly to the situation just discussed, with $m_1 = m_2$, the equations become

$$u_1 = v_1 \cos \theta_1 + v_2 \cos \theta_2 \quad \dots (i)$$

$$v_1 \sin \theta_1 = v_2 \sin \theta_2 \quad \dots (ii)$$

$$\text{and } u_1^2 = v_1^2 + v_2^2 \quad \dots (iii)$$

Squaring and adding the first two equations, we get

$$u_1^2 + v_1^2 - 2u_1 v_1 \cos \theta_1 = v_2^2$$

Combining this with the third equation, we obtain

$$2v_1^2 = 2u_1 v_1 \cos \theta_1$$

$$\text{or (since } u_1 \neq 0) \quad v_1 = u_1 \cos \theta_1 = 260 \text{ m/s}$$

From the third equation, $v_2 = 150 \text{ m/s}$

and finally from the second equation,

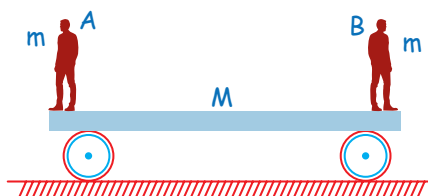
$$\theta_2 = 60^\circ.$$

The molecules move apart at right angles

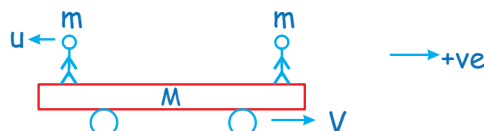
$$(\theta_1 + \theta_2 = 90^\circ).$$

Note: The student should be able to show that in an elastic collision between particles of equal mass, one of which is initially at rest, the recoiling particles always move off at right angles to one another.

- Q.** Two persons A and B, each of mass m are standing at the two ends of rail-road car of mass M . The person A jumps to the left with a horizontal speed u with respect to the car. Thereafter, the person B jumps to the right, again with the same horizontal speed u with respect to the car. Find the velocity of the car after both the persons have jumped off.



- Sol.** When person A jumps, let car achieve velocity V in forward direction. So, velocity of 'A' w.r.t. ground = $(u - V)$ in backward direction.

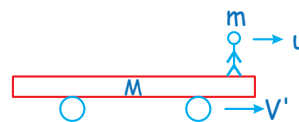


Applying momentum conservation.

$$(M + m)V - m(u - V) = 0, (M + m)V + mV = mu$$

$$V = \frac{m}{M + 2m}u$$

When person B jumps, let velocity of car becomes V' in forward direction.



So velocity of 'B' w.r.t ground = $u + V'$

Applying momentum conservation:

$$(m + M)V = MV' + m(u + V')$$

$$\frac{(m + M)mu}{M + 2m} = (m + M)V' + mu$$

$$V' = \frac{-m^2u}{(M + m)(M + 2m)}$$

- Q.** A flat car of mass M is at rest on a frictionless floor with a child of mass m standing at its edge. If child jumps off from the car towards right with an initial velocity u , with respect to the car, find the velocity of the car after its jump.

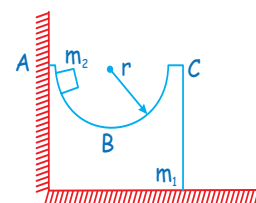
- Sol.** Let car attains a velocity v , and the net velocity of the child with respect to earth will be $u - v$, as u is its velocity with respect to car.



Initially, the system was at rest, thus according to momentum conservation, momentum after jump must be zero, as $m(u - v) = Mv$

$$v = \frac{mu}{m + M}$$

- Q.** A wedge of mass m_1 with its upper surface hemispherical in shape, as shown in Figure, rests on a smooth horizontal surface near the wall. A small block of mass m_2 slides without friction on the hemispherical surface of the wedge. What is the maximum velocity attained by the wedge?



- Sol.** As long as the block moves from A to B, the reaction on the wedge presses it to the wall. When the block reaches the lowermost position, its velocity from energy conservation is

$$v = \sqrt{2gr}$$

When the block moves along the right half of the wedge, during its upward journey as well as downward journey the reaction of the block on the wedge is towards right as shown in figure. Therefore during the entire motion of the block from B to C and C to B, the wedge is accelerated towards right. Thus to find the maximum velocity attained by the wedge at the instant when the block passes separated from the wall,

$$P_i = P_f$$

$$m_2 \sqrt{2gr} = m_1 v_1 + m_2 v_2 \quad \dots (i)$$

From energy conservation, $E_i = E_f$

$$m_2 gr = \frac{m_1 v_1^2}{2} + \frac{m_2 v_2^2}{2} \quad \dots (ii)$$

On solving eqns. (i) and (ii) simultaneously, we obtain two solutions

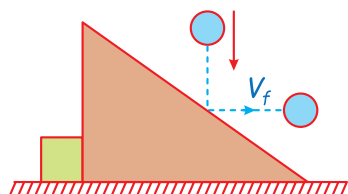
$$v_1 = 0, v_2 = \sqrt{2gr}$$

$$\text{and } v_1 = \frac{2m_2}{m_1 + m_2} (\sqrt{2gr}), v_2 = \frac{m_2 - m_1}{m_1 + m_2} (\sqrt{2gr})$$

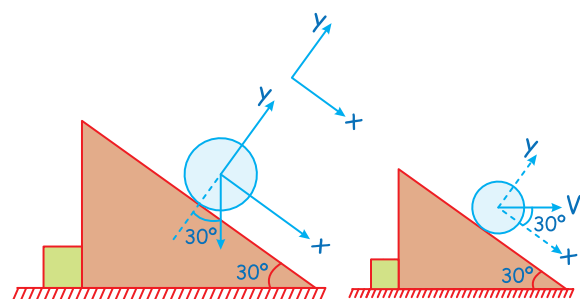
The first solution corresponds to the instant when the block reaches for the first time at point B. At this instant the block moves with velocity v_2 and the wedge is at rest. The second solution corresponds to the instant when the block has the maximum velocity

$$(v_1)_{\max} = \frac{2m_2 \sqrt{2gr}}{m_1 + m_2}$$

- Q.** As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If angle of inclination is 30° , the velocity v_f will be?



Sol. The momentum along the inclined plane (x-axis) remains conserved. i.e. $\sum p_x = 0$



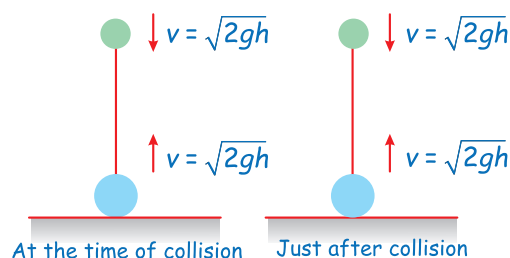
$$\Rightarrow M v_i \sin 30^\circ = M v_f \cos 30^\circ$$

$$\Rightarrow v_f = v_i \frac{\sin 30^\circ}{\cos 30^\circ} = v_i \tan 30^\circ$$

$$\Rightarrow v_f = v_i \frac{1}{\sqrt{3}} = 3 \cdot \frac{1}{\sqrt{3}} = \sqrt{3} \text{ m/s}$$

- Q.** A small ball of mass m is connected by an inextensible massless string of length l with another ball of mass $M = 4m$. They are released with zero tension in the string from a height h as shown in figure. Find the time when the string becomes taut for the first time after the mass M collides with the ground. Take all collisions to be elastic.

Sol. Just after collision velocities of both the balls will be $\sqrt{2gh}$ in opposite directions. Relative acceleration between the two balls is zero and relative velocity of approach is $2\sqrt{2gh}$. Hence they will collide after a time



At the time of collision Just after collision

$$t = \frac{l}{2\sqrt{2gh}} \quad \dots (i)$$

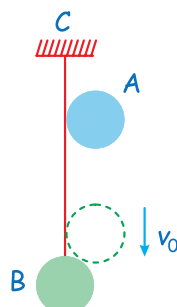
At the time of collision of two balls relative velocity of approach = $2\sqrt{2gh}$

Hence relative velocity of separation will also be $2\sqrt{2gh}$ (collision is elastic). Hence the string becomes tight after the same time

$$t = \frac{l}{2\sqrt{2gh}}$$

Hence the total time will be $2t$ or $\frac{l}{\sqrt{2gh}}$

- Q.** Ball B is hanging from an inextensible cord BC. An identical ball A is released from rest when it is just touching the cord and acquires a velocity v_0 before striking ball B. Assuming perfectly elastic impact ($e = 1$) and no friction, determine the velocity of each ball immediately after impact.

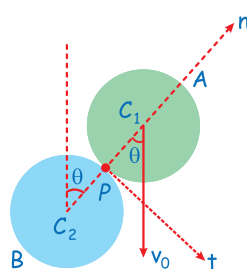


Sol. Ball A is free to move in a plane (after collision). So its velocity can be resolved in two mutually perpendicular directions. Let us resolve it along common tangent and common normal directions. Let v_t and v_n be the corresponding components of ball 'A' just after collision in these two directions. Ball B is attached to a vertical string. So, just after collision its velocity will be horizontal. Let it be v .

We have $\theta = 30^\circ$

- (i) Velocity component along common tangent direction remains unchanged. Hence,

$$u_t = v_t = v_0 \sin \theta = \frac{v_0}{2} \quad \dots (i)$$



Just before collision

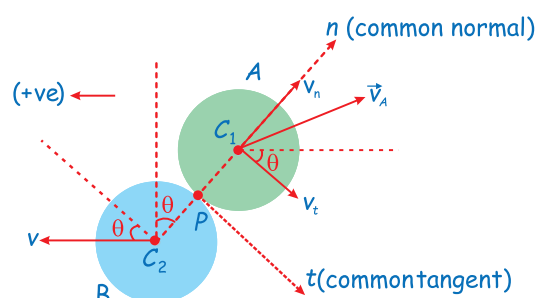
- (ii) Linear momentum along horizontal will remain unchanged. Hence,

$$mv - mv_t \cos 30^\circ - mv_n \sin 30^\circ = 0$$

$$\text{or } v - \frac{\sqrt{3}v_t}{2} - \frac{v_n}{2} = 0 \quad \dots (ii)$$

- (iii) $e = \frac{v_{\text{sep.}}}{v_{\text{app.}}} = 1$ ($\because$ elastic collision)

$$\text{i.e. } \frac{v_n + v \sin \theta}{v_0 \cos \theta} = 1$$



Just after collision

$$\therefore v_n + v \sin \theta = v_0 \cos \theta$$

$$\text{or } v_n + \frac{v}{2} = \frac{\sqrt{3}v_0}{2} \quad (\text{as } \theta = 30^\circ). \quad \dots (iii)$$

Solving Eqs. (i), (ii) and (iii), we get

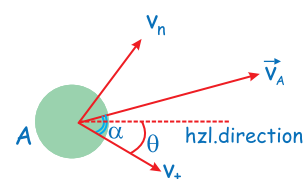
$$v = 0.693v_0, \quad v_n = 0.52v_0 \quad \text{and} \quad v_t = 0.5v_0$$

$\therefore$ Velocity of 'B' after collision $v = 0.693v_0$ (horizontally)

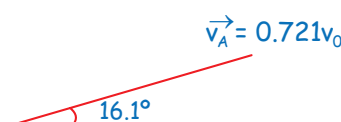
and Velocity of 'A' after collision

$$v_A = \sqrt{v_n^2 + v_t^2} = 0.721v_0$$

$$\alpha = \tan^{-1} \left(\frac{v_n}{v_t} \right) = 46.1^\circ$$



Hence, v_A makes an angle of $(46.1^\circ - 30^\circ)$ or 16.1° with horizontal as shown



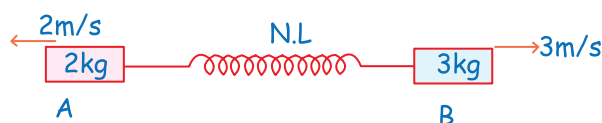
(Vcom)

$$\text{wrt com} \Rightarrow V_A V_B$$

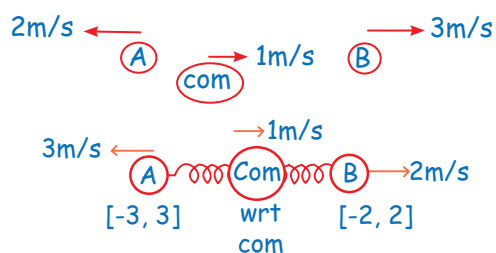
$$\text{wrt ground} \Rightarrow V_A V_B \quad (\text{Vcom add } \checkmark)$$

$\rightarrow$ jain par utre

- Q.** Find max and min velocity of A and B in entire motion



$$\text{Sol. } V_{\text{com}} = \frac{3 \times 3 - 2 \times 2}{2 + 3} = 1$$



$$\vec{V}_{A/C} = \vec{V}_A - \vec{V}_C$$

$$\vec{V}_{A/\text{ground}} = \vec{V}_{A/C} + \vec{V}_C / g$$

$$[-3 + 1, 3 + 1] \text{ wrt ground } [-2 + 1, 2 + 1]$$

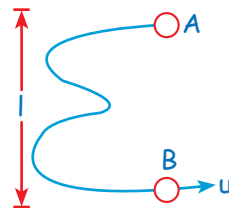
$$[-2, 4] \quad [-1, 3]$$

$$(\vec{V}_A)_{\max} = 4\hat{i}, (\vec{V}_B)_{\max} = 3\hat{i}$$

$$(\vec{V}_A)_{\min} = 0, (\vec{V}_B)_{\min} = 0$$

- Q. Two particles A and B of equal mass m each are attached by a string of length $2l$ and initially placed over a smooth horizontal table in the position shown in figure. Particle B is projected across the table with speed u perpendicular to AB as shown in figure. The ratio $\frac{v_B}{v_A}$ after the string becomes taut is $\sqrt{\frac{\alpha}{\beta}}$. Find the least integral value of $(\alpha + \beta)/2$.

(where v_B and v_A are speed of B and A respectively after the string become taut.)



Sol. $m u \cos 30^\circ = 2 \times m v_x$

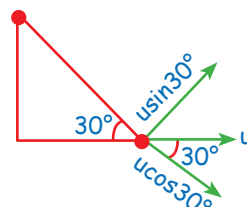
$$v_x = \frac{\sqrt{3}u}{4}$$

$$\Rightarrow v_A = \frac{\sqrt{3}u}{4}$$

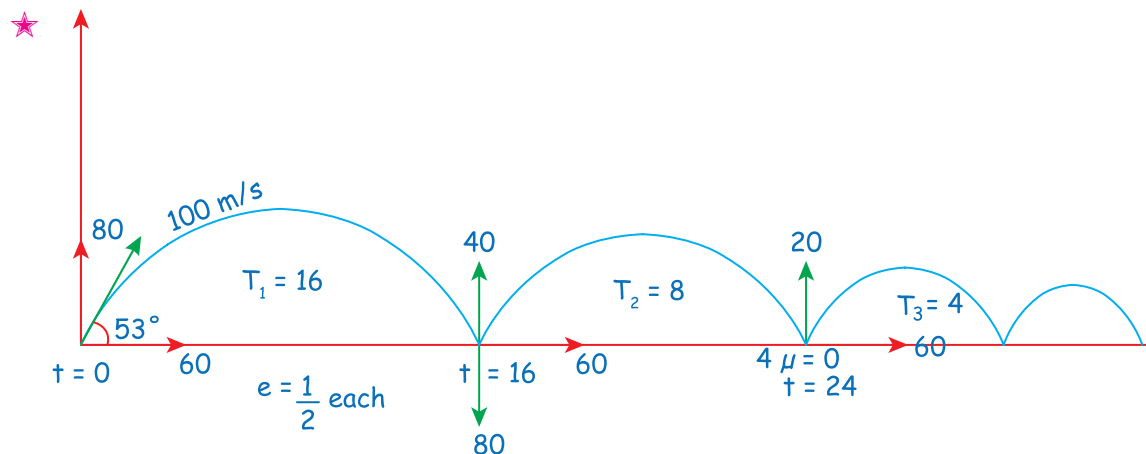
$$\Rightarrow v_B = \sqrt{\frac{3u^2}{16} + \frac{u^2}{16}} = \frac{u}{4} \times \sqrt{7}$$

$$\frac{v_B}{v_A} = \sqrt{\frac{7}{3}} = \sqrt{\frac{\alpha}{\beta}}$$

$$\frac{\alpha + \beta}{2} = 5$$



- Q. A ball is thrown at 100 m/s at 53° from horizontal. Coefficient of restitution $e = 1/2$ between ball and ground.



$$(h_2)_{\max} = \frac{(40)^2}{2g}$$

$$T_1 = 16, T_2 = 8, T_3 = 4$$